
**Road vehicles — Local Interconnect
Network (LIN) —**

**Part 8:
Electrical physical layer (EPL)
specification: LIN over DC powerline
(DC-LIN)**

Véhicules routiers — Réseau Internet local (LIN) —

*Partie 8: Spécification de couche physique électrique (EPL): LIN sur
ligne d'alimentation en courant continu (DC-LIN)*





COPYRIGHT PROTECTED DOCUMENT

© ISO 2019

All rights reserved. Unless otherwise specified, or required in the context of its implementation, no part of this publication may be reproduced or utilized otherwise in any form or by any means, electronic or mechanical, including photocopying, or posting on the internet or an intranet, without prior written permission. Permission can be requested from either ISO at the address below or ISO's member body in the country of the requester.

ISO copyright office
CP 401 • Ch. de Blandonnet 8
CH-1214 Vernier, Geneva
Phone: +41 22 749 01 11
Fax: +41 22 749 09 47
Email: copyright@iso.org
Website: www.iso.org

Published in Switzerland

Contents

Page

Foreword	iv
Introduction	v
1 Scope	1
2 Normative references	1
3 Terms, definitions, symbols, and abbreviated terms	1
3.1 Terms and definitions	2
3.2 Symbols and abbreviated terms	2
4 Electrical physical layer requirements	5
4.1 General	5
4.2 Transmitter characteristics	6
4.2.1 Transmit signal specification	6
4.2.2 Byte field modulation scheme	7
4.2.3 Break field modulation scheme	14
4.2.4 Wake-up signal modulation scheme	15
4.3 Timing requirements	18
4.3.1 Bit rate tolerance	18
4.3.2 Bit timing	19
4.3.3 Bit sample timing	19
4.3.4 TXD assert timeout event	19
4.3.5 Delay between transmitted byte field (DC-LIN node A) and received byte field (DC-LIN node A)	19
4.3.6 Delay between transmitted byte field (DC-LIN node A) and received byte field (DC-LIN node B)	19
4.4 Receiver characteristics	21
4.5 Electrical parameters	21
4.5.1 General configuration — Coupling to the DC powerline	21
4.5.2 Signal specification	22
4.5.3 Line-out monitoring	23
4.5.4 ESD compliance	28
4.6 Communication in the presence of faults	28
Annex A (informative) DC-LIN EPL peripheral interface design considerations	30
Annex B (normative) DC-LIN EPL conformance test plan	31
Bibliography	56

Foreword

ISO (the International Organization for Standardization) is a worldwide federation of national standards bodies (ISO member bodies). The work of preparing International Standards is normally carried out through ISO technical committees. Each member body interested in a subject for which a technical committee has been established has the right to be represented on that committee. International organizations, governmental and non-governmental, in liaison with ISO, also take part in the work. ISO collaborates closely with the International Electrotechnical Commission (IEC) on all matters of electrotechnical standardization.

The procedures used to develop this document and those intended for its further maintenance are described in the ISO/IEC Directives, Part 1. In particular, the different approval criteria needed for the different types of ISO documents should be noted. This document was drafted in accordance with the editorial rules of the ISO/IEC Directives, Part 2 (see www.iso.org/directives).

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. ISO shall not be held responsible for identifying any or all such patent rights. Details of any patent rights identified during the development of the document will be in the Introduction and/or on the ISO list of patent declarations received (see www.iso.org/patents).

Any trade name used in this document is information given for the convenience of users and does not constitute an endorsement.

For an explanation of the voluntary nature of standards, the meaning of ISO specific terms and expressions related to conformity assessment, as well as information about ISO's adherence to the World Trade Organization (WTO) principles in the Technical Barriers to Trade (TBT) see www.iso.org/iso/foreword.html.

This document was prepared by Technical Committee ISO/TC 22, *Road vehicles*, Subcommittee SC 31, *Data communication*.

A list of all parts in the ISO 17987 series can be found on the ISO website.

Any feedback or questions on this document should be directed to the user's national standards body. A complete listing of these bodies can be found at www.iso.org/members.html.

Introduction

ISO 17987 (all parts) specifies the use cases, communication protocol and physical layer requirements of an in-vehicle communication network called Local Interconnect Network (LIN).

The LIN protocol as proposed is an automotive-focused low speed UART-based network (Universal Asynchronous Receiver Transmitter). Some of the key characteristics of the LIN protocol are signal-based communication, schedule table-based frame transfer, master/slave communication with error detection, node configuration and diagnostic service transportation.

The LIN protocol is for low cost automotive control applications, for example door module and air conditioning systems. It serves as a communication infrastructure for low-speed control applications in vehicles by providing:

- signal-based communication to exchange information between applications in different nodes;
- bit rate support from 1 kbit/s to 20 kbit/s;
- deterministic schedule table-based frame communication;
- network management that wakes up and puts the LIN cluster into sleep mode in a controlled manner;
- status management that provides error handling and error signalling;
- transport layer that allows large amount of data to be transported (such as diagnostic services);
- specification of how to handle diagnostic services;
- electrical physical layer specifications;
- node description language describing properties of slave nodes;
- network description file describing behaviour of communication;
- application programmer's interface.

ISO 17987 (all parts) is based on the open systems interconnection (OSI) basic reference model as specified in ISO/IEC 7498-1 which structures communication systems into seven layers.

The OSI model structures data communication into seven layers called (top down) application layer (layer 7), presentation layer, session layer, transport layer, network layer, data link layer and physical layer (layer 1). A subset of these layers is used in ISO 17987 (all parts).

ISO 17987 (all parts) distinguishes between the services provided by a layer to the layer above it and the protocol used by the layer to send a message between the peer entities of that layer. The reason for this distinction is to make the services, especially the application layer services and the transport layer services, reusable also for other types of networks than LIN. In this way, the protocol is hidden from the service user and it is possible to change the protocol if special system requirements demand it.

ISO 17987 (all parts) provides all documents and references required to support the implementation of the requirements related to the following.

- ISO 17987-1: This part provides an overview of the ISO 17987 (all parts) and structure along with the use case definitions and a common set of resources (definitions, references) for use by all subsequent parts.
- ISO 17987-2: This part specifies the requirements related to the transport protocol and the network layer requirements to transport the PDU of a message between LIN nodes.
- ISO 17987-3: This part specifies the requirements for implementations of the LIN protocol on the logical level of abstraction. Hardware related properties are hidden in the defined constraints.

- ISO 17987-4: This part specifies the requirements for implementations of active hardware components which are necessary to interconnect the protocol implementation.
- ISO/TR 17987-5: This part specifies the LIN application programmers interface (API) and the node configuration and identification services. The node configuration and identification services are specified in the API and define how a slave node is configured and how a slave node uses the identification service.
- ISO 17987-6: This part specifies tests to check the conformance of the LIN protocol implementation according to ISO 17987-2 and ISO 17987-3. This comprises tests for the data link layer, the network layer and the transport layer.
- ISO 17987-7: This part specifies tests to check the conformance of the LIN electrical physical layer implementation (logical level of abstraction) according to ISO 17987-4.
- ISO 17987-8: This part specifies the requirements for implementations of the DC powerline electrical physical layer (EPL) for the LIN communications system as well as a conformance test plan for the EPL.

Road vehicles — Local Interconnect Network (LIN) —

Part 8:

Electrical physical layer (EPL) specification: LIN over DC powerline (DC-LIN)

1 Scope

This document specifies an additional electrical physical layer (EPL) for the Local Interconnect Network (LIN) of the ISO 17987 series. It specifies the transmission over DC powerline without affecting the LIN higher layers, hereafter named DC-LIN.

The DC-LIN EPL uses a high-frequency modulated carrier to propagate UART bytes (byte-oriented) over the DC powerline.

This document specifies the electrical characteristics, the modulation method of the transmission, and how to impose the carrier signal on the DC powerlines.

The DC-LIN EPL supports bit rates of 9 615 bit/s, 10 417 bit/s, and 19 230 bit/s.

The DC-LIN EPL is applicable for a wide range of DC powerlines including 12-V and 24-V operations, allowing communicating between different DC powerlines via a coupling capacitor. A DC-LIN EPL interface to powerline example is described in [Annex A](#).

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 17987-4:2016, *Road vehicles — Local Interconnect Network (LIN) — Part 4: Electrical physical layer (EPL) specification 12 V/24 V*

ISO 17987-6, *Road vehicles — Local Interconnect Network (LIN) — Part 6: Protocol conformance test specification*

IEC 61000-4-2, *Electromagnetic compatibility (EMC) - Part 4-2: Testing and measurement techniques - Electrostatic discharge immunity test*

3 Terms, definitions, symbols, and abbreviated terms

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- ISO Online browsing platform: available at <https://www.iso.org/obp>
- IEC Electropedia: available at <http://www.electropedia.org/>

3.1 Terms and definitions

3.1.1

BR_9_6K

DC-LIN EPL operating at nominal bit rate of 9 615 bit/s

3.1.2

BR_10K

DC-LIN EPL operating at nominal bit rate of 10 417 bit/s

3.1.3

BR_19_2K

DC-LIN EPL operating at nominal bit rate of 19 230 bit/s

3.1.4

byte field

byte that consists of one start bit, eight data bits, and one stop bit

3.1.5

byte field sync preamble

sequence of phase shifts at the beginning of EPL byte field modulation used for byte synchronization

3.1.6

carrier frequency

DC-LIN EPL center frequency that is altered (modulated) to transfer data

3.1.7

coupling capacitor

capacitor for blocking the DC powerline voltage to/from a DC-LIN EPL

3.1.8

DC-LIN EPL RX mode

mode that DC-LIN EPL de-asserts line-out and controls RXD according to signal at line-in

3.1.9

DC-LIN EPL TX mode

mode that DC-LIN EPL controls line-out according to logic state present at TXD

3.1.10

start bit

logic low ('0') of the first bit of a byte field

3.1.11

stop bit

logic high ('1') of the last bit of a byte field

3.2 Symbols and abbreviated terms

'0' logical 0

'1' logical 1

AC alternate current

API application programmers interface

B_{data_bit} byte field data bit signalled on RXD at DC-LIN EPL receiver side

B_{data_bit_local} byte field data bit signalled on RXD at DC-LIN EPL transmitter side

B_{del}	break delimiter signalled on RXD at DC-LIN EPL receiver side
B_{del_local}	break delimiter signalled on RXD at DC-LIN EPL transmitter side
$B_{err_data_bit_local}$	byte field error data bit signalled on RXD at DC-LIN EPL transmitter side
$B_{fe_stop_bit}$	byte field frame error stop bit signalled on RXD at DC-LIN EPL receiver side
$B_{fe_stop_bit_local}$	byte field frame error stop bit signalled on RXD at DC-LIN EPL transmitter side
$B_{fe_del_local}$	break field delimiter frame error signalled on RXD at DC-LIN EPL transmitter side
$B_{fe_start_bit_local}$	byte field frame error start bit signalled on RXD at DC-LIN EPL transmitter side
BR	DC-LIN EPL operating bit rate
B_{start_bit}	byte field start bit signalled on RXD at DC-LIN EPL receiver side
$B_{start_bit_local}$	byte field start bit signalled on RXD at DC-LIN EPL transmitter side
B_{stop_bit}	byte field stop bit signalled on RXD at DC-LIN EPL receiver side
$B_{stop_bit_local}$	byte field stop bit signalled on RXD at DC-LIN EPL transmitter side
CB_{TX}	consecutive byte field transmission
CF_{max}	maximal number of carrier frequencies implemented in IUT transmit signal
DC	direct current
EPL	electrical physical layer
ESD	electrostatic discharge
fc_i	carrier frequency
FB_{TX}	first byte field transmission
I	in-phase signal component
IUT	implementation under test
bit/s	bit per second
LIN	Local Interconnect Network
line-in	modulated carrier signal input pin to the EPL from the DC powerline
line-out	modulated carrier signal output pin from the EPL to the DC powerline
LT	lower tester
L_{out_lev}	line-out sampled monitoring level
$L_{out_thr_lev}$	line-out monitoring peak threshold level
$L_{out_err_cond}$	line-out monitoring error condition
max.	maximum
min.	minimum

$P_{\text{byte_length}}$	nominal length of modulated byte field in t_{BIT}
$P_{\text{data_bit_len}}$	nominal length of modulated byte field data bit
P_{data}	data bit modulation phases
$P_{\text{dh_1}}$	data bit '1' 1 st phase shift
$P_{\text{dh_2}}$	data bit '1' 2 nd phase shift
$P_{\text{dh_3}}$	data bit '1' 3 rd phase shift
$P_{\text{dl_1}}$	data bit '0' 1 st phase shift
$P_{\text{dl_2}}$	data bit '0' 2 nd phase shift
$P_{\text{dl_3}}$	data bit '0' 3 rd phase shift
$P_{\text{ref_consec}}$	reference phase of consecutive byte transmission
$P_{\text{ref_first}}$	reference phase of the first byte transmission
$P_{\text{sync_p}}$	sync preamble modulation phases
$P_{\text{sp_1}}$	sync preamble 1 st phase shift
$P_{\text{sp_2}}$	sync preamble 2 nd phase shift
$P_{\text{sp_3}}$	sync preamble 3 rd phase shift
$P_{\text{sp_4}}$	sync preamble 4 th phase shift
$P_{\text{sp_5}}$	sync preamble 5 th phase shift
$P_{\text{sp_6}}$	sync preamble 6 th phase shift
$P_{\text{sp_7}}$	sync preamble 7 th phase shift
$P_{\text{sp_8}}$	sync preamble 8 th phase shift
$P_{\text{sp_9}}$	sync preamble 9 th phase shift
$P_{\text{sp_10}}$	sync preamble 10 th phase shift
$P_{\text{sp_11}}$	sync preamble 11 th phase shift
$P_{\text{sp_12}}$	sync preamble 12 th phase shift
$P_{\text{sp_13}}$	sync preamble 13 th phase shift
$P_{\text{sp_14}}$	sync preamble 14 th phase shift
$P_{\text{sp_15}}$	sync preamble 15 th phase shift
$P_{\text{sp_16}}$	sync preamble 16 th phase shift
$P_{\text{sp_17}}$	sync preamble 17 th phase shift
$P_{\text{sp_18}}$	sync preamble 18 th phase shift
$P_{\text{sync_pre_len}}$	nominal length of modulated byte field sync preamble

Q	quadrature signal component
R_{rx_mode}	DC-LIN EPL RX mode
RX_{err_cond}	DC-LIN EPL RX node error condition
t_{BIT}	bit time
t_{break_field}	break field data bits '0' length in t_{BIT}
$t_{l_out_mon_l}$	line-out monitoring active duration
$t_{l_out_mon_samp}$	line-out monitoring sample time
$t_{l_out_mon_start}$	start time of line-out monitoring
t_{RX_delay}	delay between transmitted byte field over to the DC powerline and the receiver reconstructed byte field from the DC powerline
$t_{RX_delay_local}$	delay between transmitted byte field on TXD and the received byte field on RXD at a DC-LIN EPL node (locally)
$t_{rx_proc_max}$	maximum process time at RX DC-LIN EPL side
$t_{rx_proc_min}$	minimum process time at RX DC-LIN EPL side
t_{SB_TX}	start time of byte field sync preamble transmission on line-out
T_{tx_mode}	DC-LIN EPL TX mode
$t_{txd_min_assert}$	minimum TXD assert ('0') time without timeout event (deactivating T_{tx_mode})
$t_{txd_min_recover}$	minimum TXD deassert ('1') time after timeout event (T_{tx_mode} remains deactivated)
$\overline{TXD}$	logical inverted TXD (i.e. '0' becomes '1' and vice versa)
typ.	typical
UT	upper tester
V_{pp}	volt peak-to-peak
V_{PWL_max}	maximum rating for the DC powerline
φ	carrier phases
$\Delta f c_i$	carrier frequency resolution
°	degree

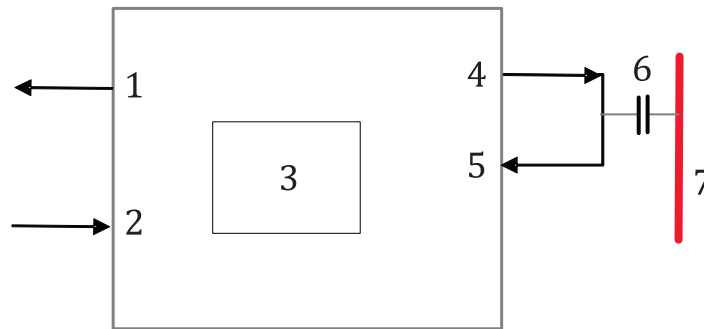
4 Electrical physical layer requirements

4.1 General

The DC-LIN EPL is the physical media access sub-layer, which links the data link layer as standardized in ISO 17987-3 and the DC powerlines (physical medium dependent sub-layer). [Figure 1](#) depicts an example of a DC-LIN EPL. The EPL consists of a modem, which encodes the data from the data link layer into the modulated carrier signal that is coupled to the DC powerline. The modem also decodes the received data on the DC powerline and provides this to the data link layer at the receivers. Follow the DC-LIN EPL conformance test plan in [Annex B](#).

The DC-LIN EPL transmitter encodes each byte field sent by the data link layer on TXD into a modulated carrier signal, which is transferred over the DC powerline. The modulation consists of a predefined combination of phase shifts according to the byte field modulation scheme specified in [4.2.2](#).

The DC-LIN EPL receiver decodes the received modulated byte from the DC powerline and signals it on RXD to the data link layer.



Key

- 1 RXD receive data pin
- 2 TXD transmit data pin
- 3 modem
- 4 line-out
- 5 line-in
- 6 coupling capacitor
- 7 DC powerline

Figure 1 — Example of a DC-LIN EPL

4.2 Transmitter characteristics

4.2.1 Transmit signal specification

The transmit signal shall be constructed as the sum of up to four (redundant) selectable modulated carrier frequencies.

The definition of the transmit signal is given in [Formula \(1\)](#).

Definition of the formula:

$$\text{Transmit signal} = \sum_{i=1}^n A \times \cos[(360^\circ \times fc_i \times t) + \varphi(t)] \quad (1)$$

where

- A is the gain amplitude of the carrier signal in volts;
- t is the time in seconds;
- fc_i is the selected carrier frequency;
- $\varphi(t)$ is the carrier frequency phase; $\varphi(t) = 0^\circ, 90^\circ, 180^\circ, 270^\circ$;
 $\varphi(t)$ changes as a function of the byte field modulation scheme;
- n is the maximal carrier frequency selection per transmit signal;
 $n = 1, 2, 3, 4$.

[Table 1](#) specifies the DC-LIN EPL carrier frequency.

Table 1 — Carrier frequency specification

Parameter	Description	Min.	Max.	Unit	Accuracy
fc_i	Carrier frequency band	5	30	MHz	$\pm 0,02 \%$
Δfc_i	Carrier frequency resolution	0,1	0,1	MHz	

In essence, $fc_i \in \{5 \text{ MHz}; 5,1 \text{ MHz}; 5,2 \text{ MHz}; \dots 29,8 \text{ MHz}; 29,9 \text{ MHz}; 30 \text{ MHz}\}$.

4.2.2 Byte field modulation scheme

4.2.2.1 Byte field modulation scheme structure

A byte field modulation scheme shall consist of a byte field sync preamble modulation (specified in [4.2.2.3](#)) and a byte field data bit modulation (specified in [4.2.2.4](#)). A first byte field modulation shall start with a dedicated reference phase (specified in [4.2.2.2](#)).

A byte field start bit and stop bit shall not be included in a byte field modulation scheme. At receiving nodes, the DC-LIN EPL shall reconstruct both the start bit and stop bit artificially on RXD (see [4.4](#)).

[Figure 2](#) depicts a byte field modulation scheme structure.

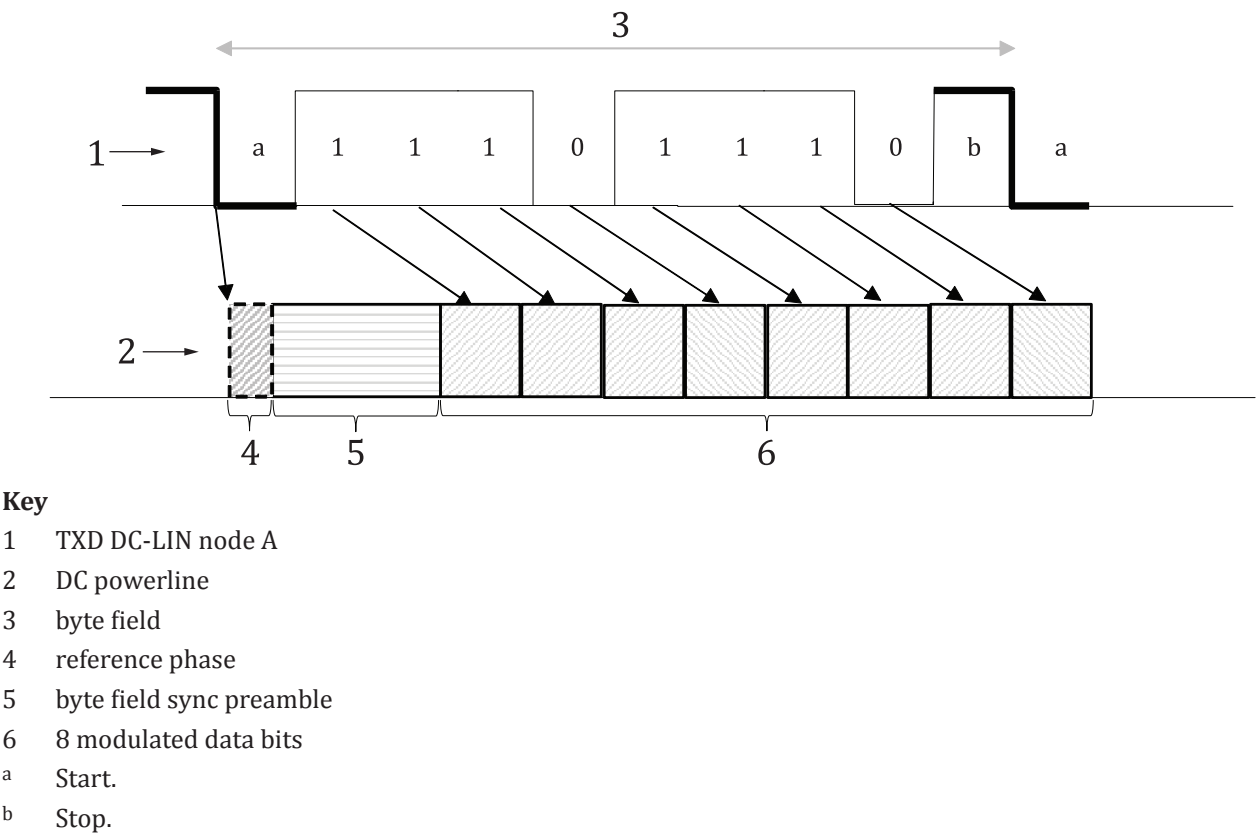


Figure 2 — Byte field modulation scheme structure

The byte field modulation scheme shall consist of a sequence of $\pm 90^\circ$ phase shifts while $L_{\text{out_err_cond}}$ is inactive (see 4.2.2.3 and 4.2.2.4).

While $L_{\text{out_err_cond}}$ is active, and only after completion of byte field sync preamble transmission, the transmit signal shall consist of no phase shifts (i.e. constant phase transmission) for the remaining field transmission time (see 4.5.3).

Figure 3 specifies the transmitting phases for 0° , 90° , 180° , and 270° [i.e. $\varphi(t)$].

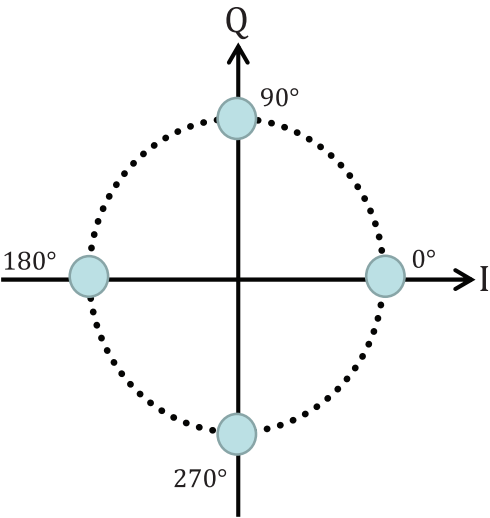


Figure 3 — Transmitter phase's definition

4.2.2.2 Reference phase

The DC-LIN EPL shall transmit a reference phase ($P_{\text{ref_first}}$) prior to the first phase transmission of a byte field sync preamble modulation.

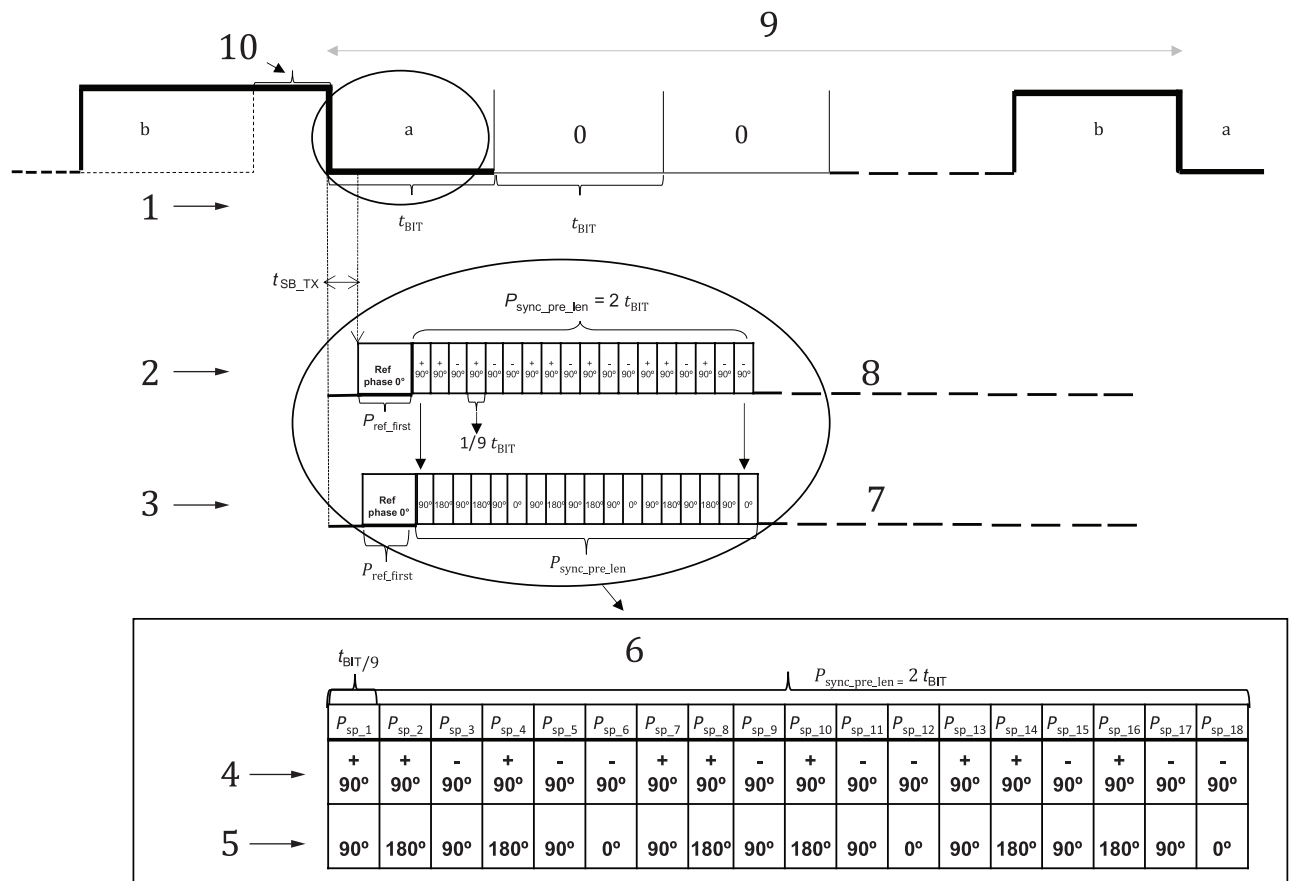
A first byte field transmission (FB_{TX}) shall be interpreted as a start of a byte field transmission by the data link layer with an inter-byte space longer than $1/3 t_{\text{BIT}}$.

A consecutive byte field transmission (CB_{TX}) shall be interpreted as a start of a byte field transmission by the data link layer with an inter-byte space no longer than $1/3 t_{\text{BIT}}$.

In the case of a FB_{TX} , a dedicated reference phase shall be transmitted ($P_{\text{ref_first}}$) as specified in Figure 4.

In the case of a CB_{TX} (i.e. not a FB_{TX}), the last phase of the previous byte field modulation transmission shall be used as the reference phase ($P_{\text{ref_consec}}$) of the next byte field sync preamble modulation (as specified in Figure 5).

Figure 4 shows an example of byte field sync preamble modulation of a first byte field transmission.



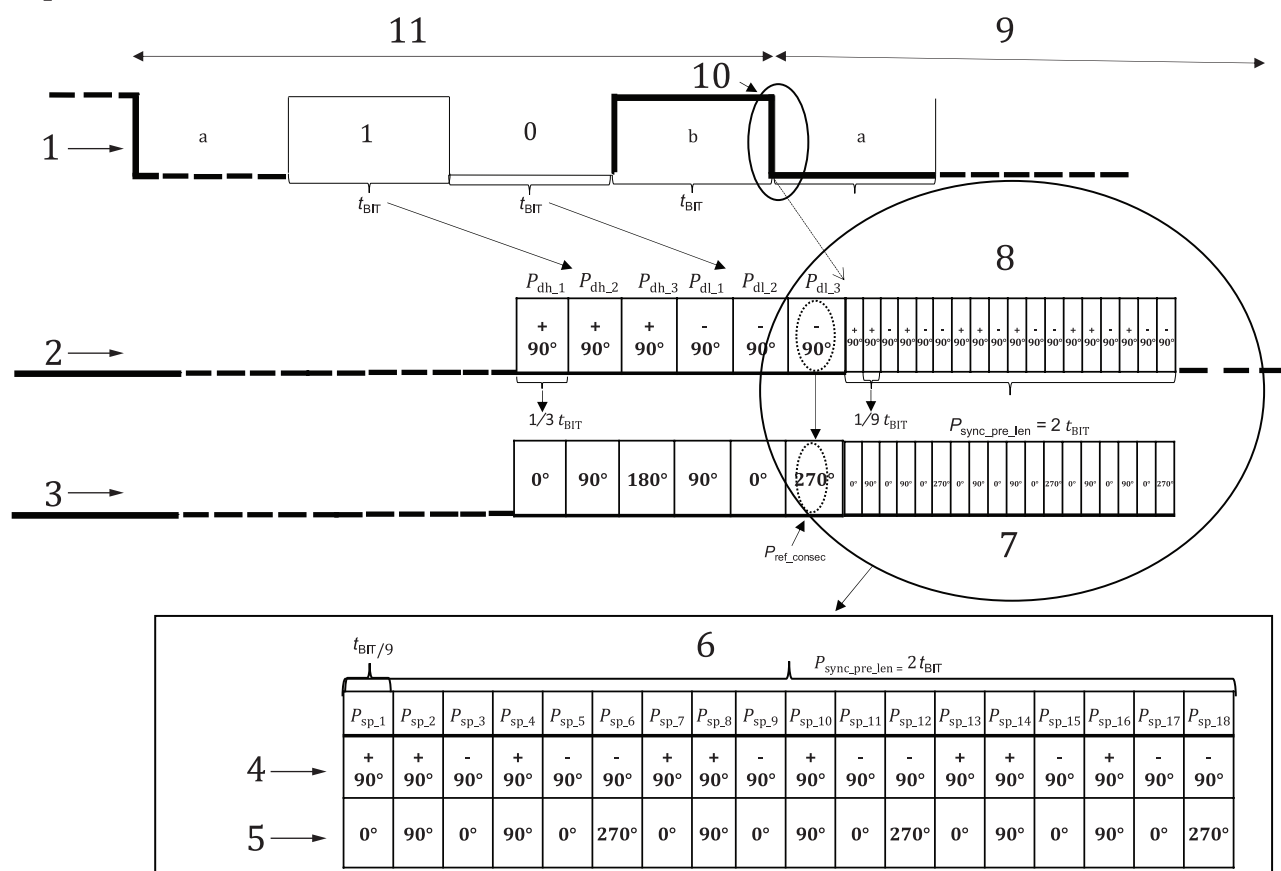
Key

- | | | | |
|---|--------------------------|----|--|
| 1 | TXD DC-LIN node A | 7 | byte field sync preamble - actual TX phases |
| 2 | DC powerline | 8 | byte field sync preamble - phase shifts |
| 3 | TX phases | 9 | DC-LIN first byte field transmission of node A |
| 4 | phase shifts | 10 | Inter-byte space > $1/3 t_{\text{BIT}}$ |
| 5 | actual TX phases | a | Start. |
| 6 | byte field sync preamble | b | Stop. |

Figure 4 — Example of byte field sync preamble modulation of first byte field TX

Figure 5 shows an example of byte field sync preamble modulation of consecutive byte field transmissions.

P_{ref_consec} is the last phase of the previous byte field transmission.



Key

- | | | | |
|---|--------------------------------------|----|---|
| 1 | TXD DC-LIN node A | 8 | byte field sync preamble – phase shifts |
| 2 | DC powerline | 9 | consecutive byte field transmission |
| 3 | example of TX phases | 10 | Inter-byte space < $1/3 t_{BIT}$ |
| 4 | phase shifts | 11 | first byte field transmission |
| 5 | example of TX phases | a | Start. |
| 6 | byte field sync preamble | b | Stop. |
| 7 | byte field sync preamble – TX phases | | |

Figure 5 — Example of byte field sync preamble modulation of consecutive byte field TX

Table 2 specifies the reference phase length and value.

Table 2 — Reference phase length and value definition

Parameter	Description	Phase	Min. phase length	Max. phase length	Unit
P_{ref_first}	Reference phase of a first byte field transmission (shown in Figure 4)	0°	2/9	1/3	t_{BIT}
P_{ref_consec}	Reference phase of a consecutive byte field transmission (shown in Figure 5)	Last byte field transmitted phase (0°/90°/180°/270°)	1/3	2/3	t_{BIT}

4.2.2.3 Byte field sync preamble modulation

Upon detection of falling edge of a start bit (t_{SB_TX}) from the data link layer, a byte field sync preamble modulation transmission shall start on line-out.

[Table 3](#) specifies the start time of a byte field sync preamble transmission.

Table 3 — Start time of byte field sync preamble transmission definition

Parameter	Description	Min.	Typ.	Max.	Unit
t_{SB_TX}	Start time of byte field sync preamble transmission on line-out (shown in Figure 4)	—	—	2/16	t_{BIT}

The byte field sync preamble modulation shall consist of 18 phase shifts.

The nominal length of modulated byte field sync preamble ($P_{sync_pre_len}$) is given in [Formula \(2\)](#).

Definition of the formula:

$$P_{sync_pre_len} = 2 t_{BIT} \quad (2)$$

[Table 4](#) specifies the byte field sync preamble phase shifts.

Table 4 — Byte field sync preamble phase shifts definition

Parameter	Description	Phase shift	Phase length
P_{sp_1}	Sync preamble 1 st phase shift	+90°	1/9 t_{BIT} (shown in Figure 4 and Figure 5)
P_{sp_2}	Sync preamble 2 nd phase shift	+90°	
P_{sp_3}	Sync preamble 3 rd phase shift	−90°	
P_{sp_4}	Sync preamble 4 th phase shift	+90°	
P_{sp_5}	Sync preamble 5 th phase shift	−90°	
P_{sp_6}	Sync preamble 6 th phase shift	−90°	
P_{sp_7}	Sync preamble 7 th phase shift	+90°	
P_{sp_8}	Sync preamble 8 th phase shift	+90°	
P_{sp_9}	Sync preamble 9 th phase shift	−90°	
P_{sp_10}	Sync preamble 10 th phase shift	+90°	
P_{sp_11}	Sync preamble 11 th phase shift	−90°	
P_{sp_12}	Sync preamble 12 th phase shift	−90°	
P_{sp_13}	Sync preamble 13 th phase shift	+90°	
P_{sp_14}	Sync preamble 14 th phase shift	+90°	
P_{sp_15}	Sync preamble 15 th phase shift	−90°	
P_{sp_16}	Sync preamble 16 th phase shift	+90°	
P_{sp_17}	Sync preamble 17 th phase shift	−90°	
P_{sp_18}	Sync preamble 18 th phase shift	−90°	

4.2.2.4 Byte field data bit modulation

Upon completion of the byte field sync preamble transmission period ($P_{sync_pre_len}$), the DC-LIN EPL shall transmit a byte field data bit modulation, bit-by-bit, for each one of the eight data bits transmitted from the data link layer.

The byte field data bit modulation shall consist of three phase shifts.

The nominal length of a modulated byte field data bit ($P_{data_bit_len}$) is given in [Formula \(3\)](#).

Definition of the formula:

$$P_{data_bit_len} = 1\ t_{BIT}$$

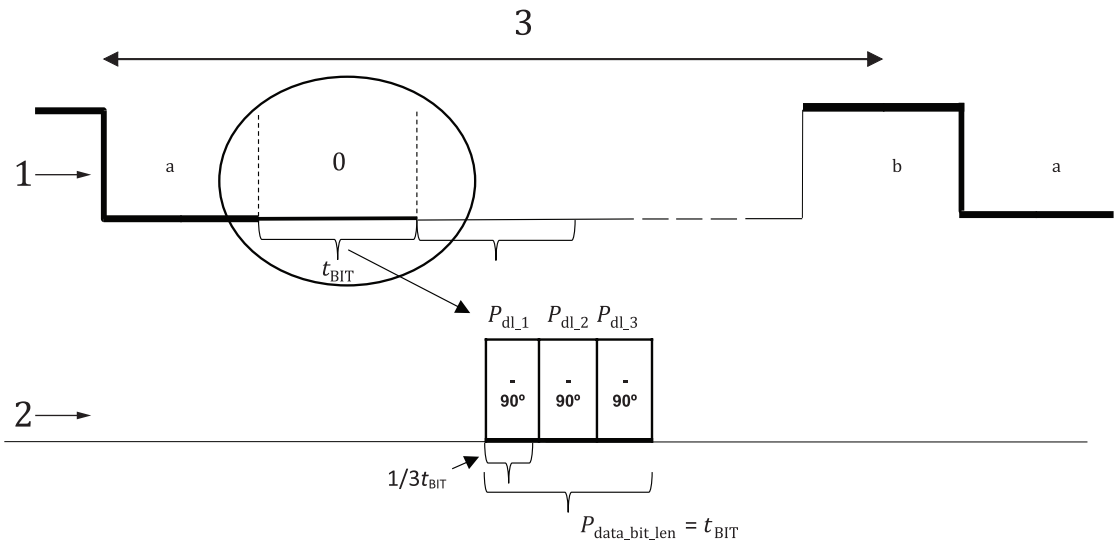
(3)

[Table 5](#) specifies the phase shift of the byte field data bit modulation.

Table 5 — Byte field data bit phase shift modulation definition

Data bit value	Parameter	Description	Phase shift	Phase length	Condition/comment
1	P_{dh_1}	1 st phase shift	+90°	$1/3\ t_{BIT}$	$L_{out_err_cond}$ is inactive (see 4.5.3)
	P_{dh_2}	2 nd phase shift	+90°		
	P_{dh_3}	3 rd phase shift	+90°		
0	P_{dl_1}	1 st phase shift	-90°		
	P_{dl_2}	2 nd phase shift	-90°		
	P_{dl_3}	3 rd phase shift	-90°		

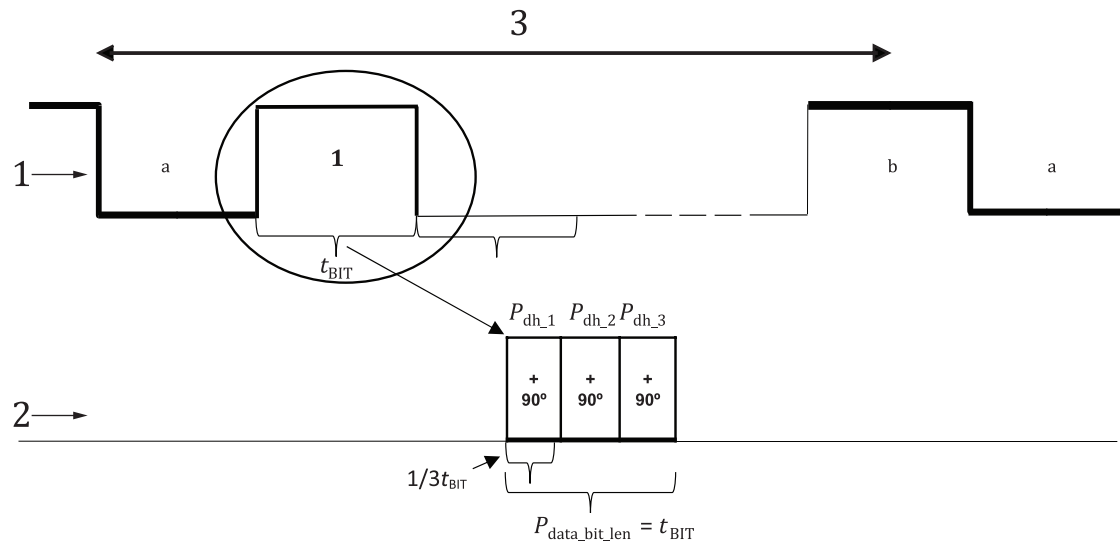
[Figure 6](#) shows a bit modulation example of data bit ‘0’.



- Key
- 1 TXD
 - 2 DC powerline
 - 3 byte field
 - a Start.
 - b Stop.

Figure 6 — Example of bit ‘0’ modulation

[Figure 7](#) shows a bit modulation example of data bit ‘1’.



Key

- 1 TXD
- 2 DC powerline
- 3 byte field
- a Start.
- b Stop.

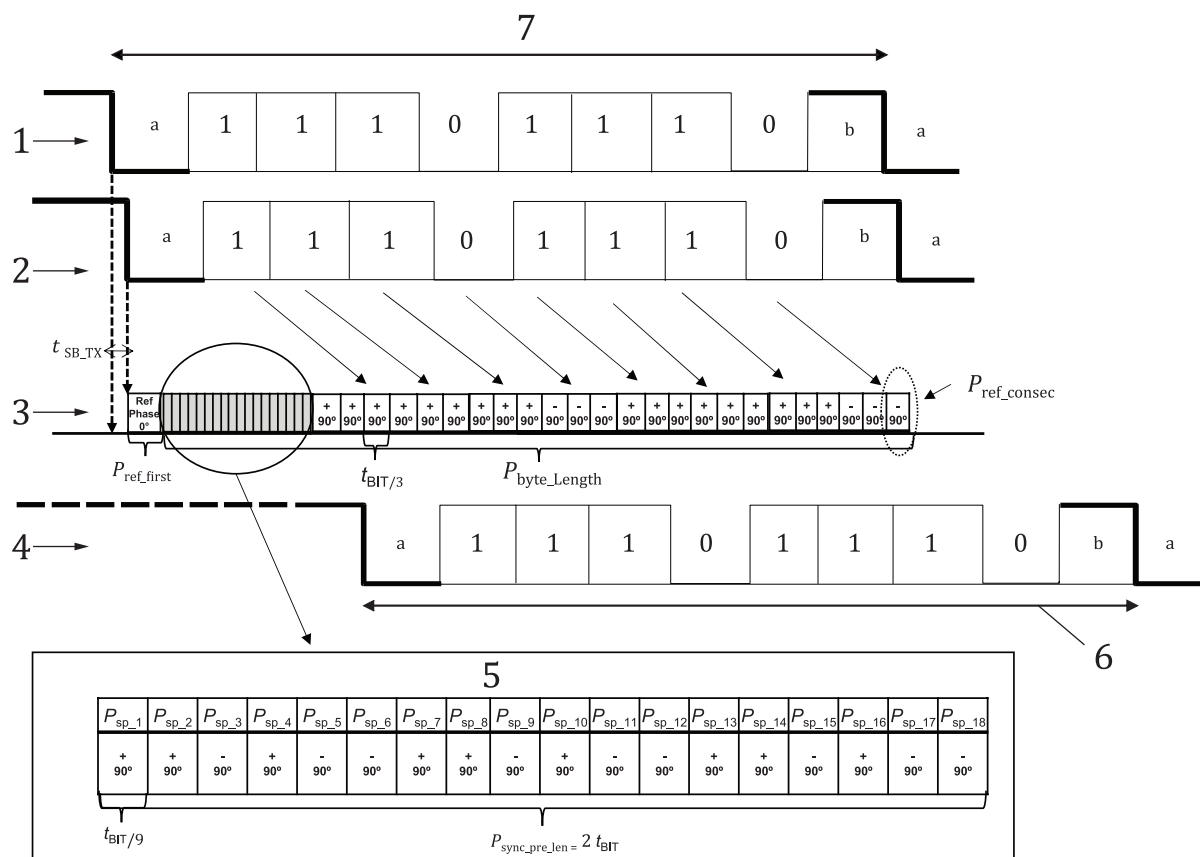
Figure 7 — Example of data bit '1' modulation

The nominal length of one modulated byte ($P_{\text{byte_length}}$) is given in [Formula \(4\)](#).

Definition of the formula:

$$P_{\text{byte_length}} = P_{\text{sync_pre_len}} + 8 \times P_{\text{data_bit_len}} = 10 t_{\text{BIT}} \quad (4)$$

[Figure 8](#) shows an example of modulation scheme of data byte 01110111₂.



Key

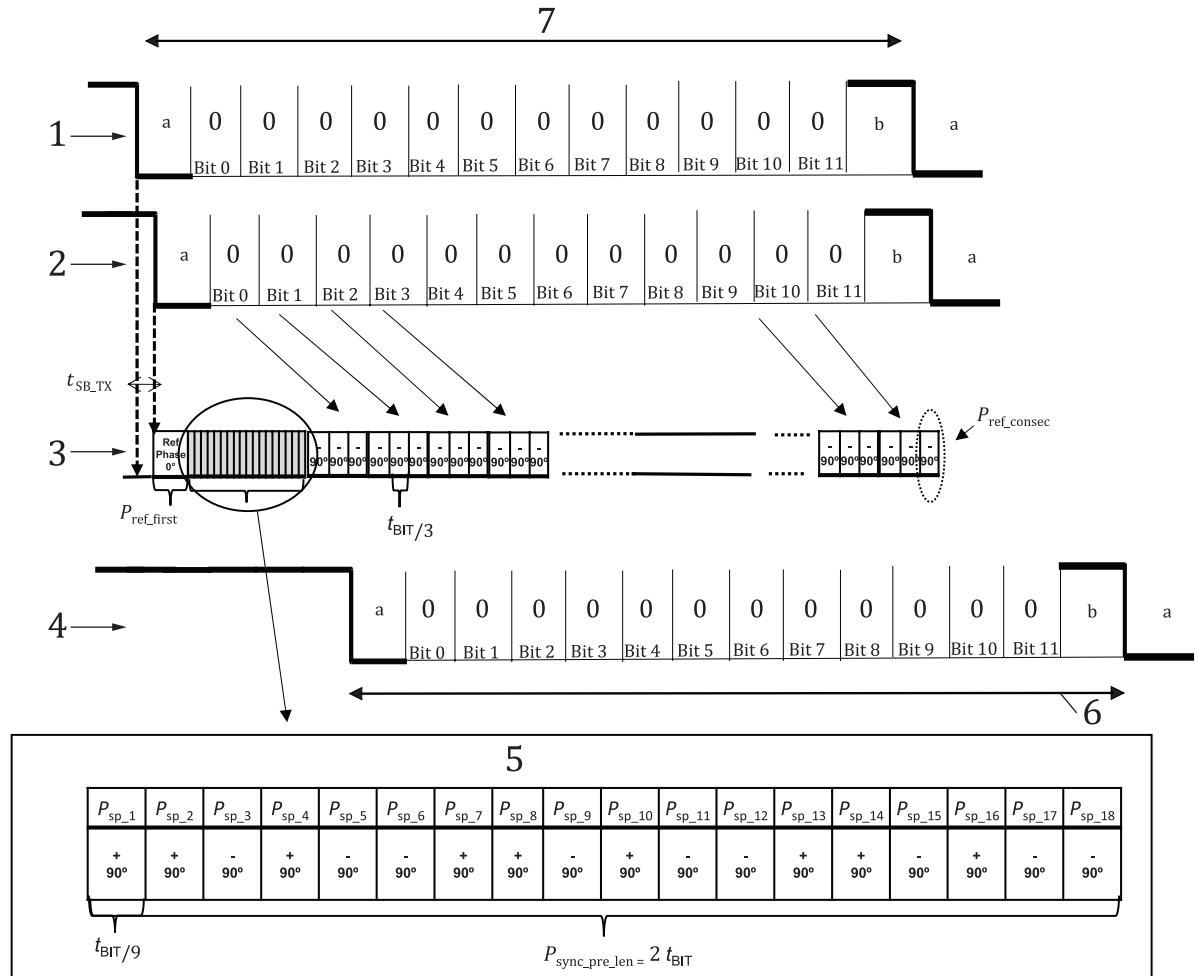
- 1 TXD DC-LIN node A
- 2 RXD DC-LIN node A
- 3 DC powerline
- 4 RXD DC-LIN node B
- 5 byte field sync preamble
- 6 DC-LIN byte field reception of node B
- 7 DC-LIN byte field transmission of node A
- a Start.
- b Stop.

Figure 8 — Example of a byte modulation scheme between a transmitting node A and receiving node B

4.2.3 Break field modulation scheme

The DC-LIN EPL shall transmit the break field over the DC powerline with the number of bits of the detected break field on TXD. The break field modulation scheme is the same as the byte field modulation scheme, but all data bits are '0's.

[Figure 9](#) shows an example of break field modulation transmission.



Key

- 1 TXD DC-LIN node A
- 2 RXD DC-LIN node A
- 3 DC powerline
- 4 RXD DC-LIN node B
- 5 byte field sync preamble
- 6 DC-LIN break field reception of node B
- 7 DC-LIN break field transmission of node A
- a Start.
- b Break delimiter.

Figure 9 — Example of a break field modulation transmission between a transmitting node A and receiving node B

4.2.4 Wake-up signal modulation scheme

The wake-up signal modulation is subject to the wake-up signal length on TXD (as specified in ISO 17987-2:2016, 5.3.1) and shall be transmitted either as a byte field modulation or as a break field modulation.

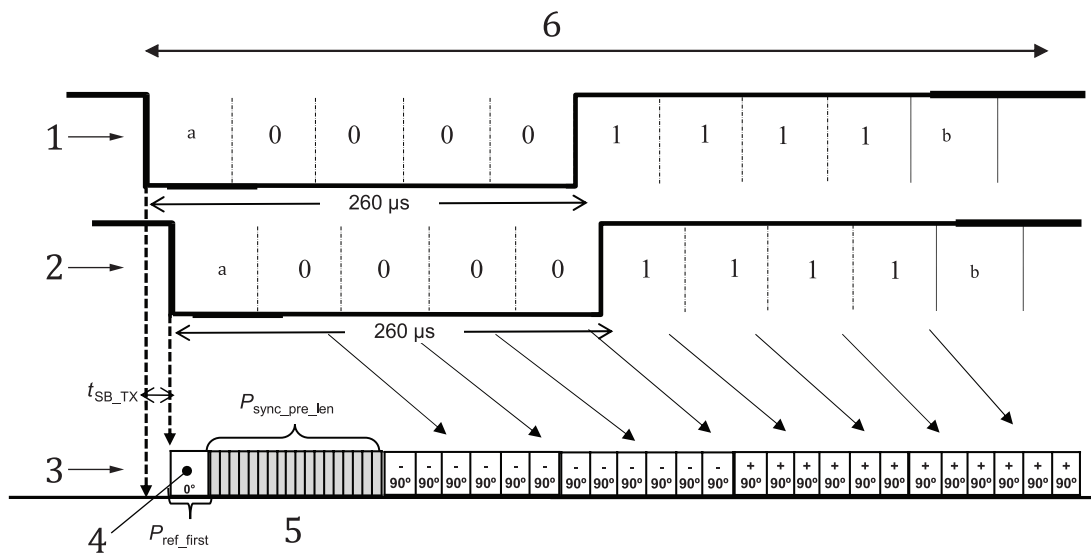
When the wake-up signal length on TXD is at least 10 t_{BIT} (i.e. a break field), the wake-up signal modulation shall be transmitted as a break field modulation scheme (see 4.2.3).

When the wake-up signal length on TXD is less than $10 t_{BIT}$, the wake-up signal modulation shall be transmitted as a byte field modulation scheme (see 4.2.2.4) consisting of ‘0’ and ‘1’ data bit modulation (i.e. the wake-up signal is padded with ‘1’ data bit modulation to a complete byte field modulation transmission).

Figures 10, 11, and 12 show examples of a wake-up signal modulation transmission by a DC-LIN EPL that is set to BR_19_2K.

EXAMPLE Figure 10 depicts a wake-up signal with the length of 260 μ s. The wake-up signal is encoded by the DC-LIN EPL to a start bit, four ‘0’ data bits, four ‘1’ data bits and a stop bit (i.e. F0₁₆).

The wake-up signal modulation is similar to a F0₁₆ byte field modulation.

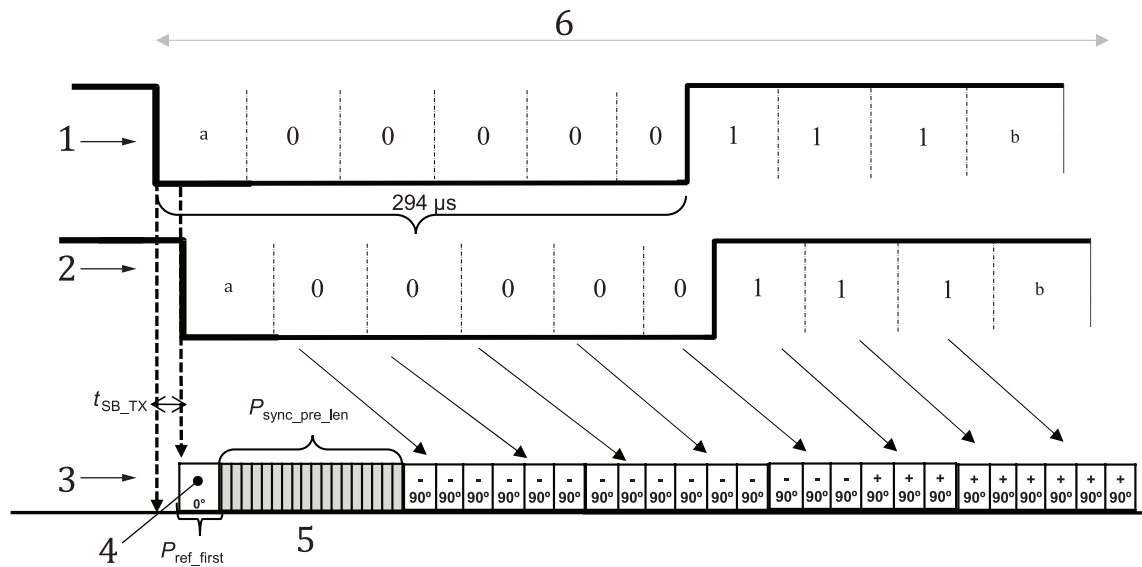


- Key**
- 1 TXD wake-up signal DC-LIN node A
 - 2 RXD wake-up signal DC-LIN node A
 - 3 DC powerline
 - 4 ref phase 0°
 - 5 byte field sync preamble
 - 6 DC-LIN wake-up signal transmission of node A
 - a Start.
 - b Stop.

Figure 10 — Example of a wake-up signal modulation similar to byte field modulation

EXAMPLE Figure 11 depicts a wake-up signal with the length of 294 μ s. The wake-up signal is encoded by the DC-LIN EPL to a start bit, five ‘0’ data bits, three ‘1’ data bits and a stop bit (i.e. E0₁₆).

The wake-up signal modulation is similar to a E0₁₆ byte field modulation.



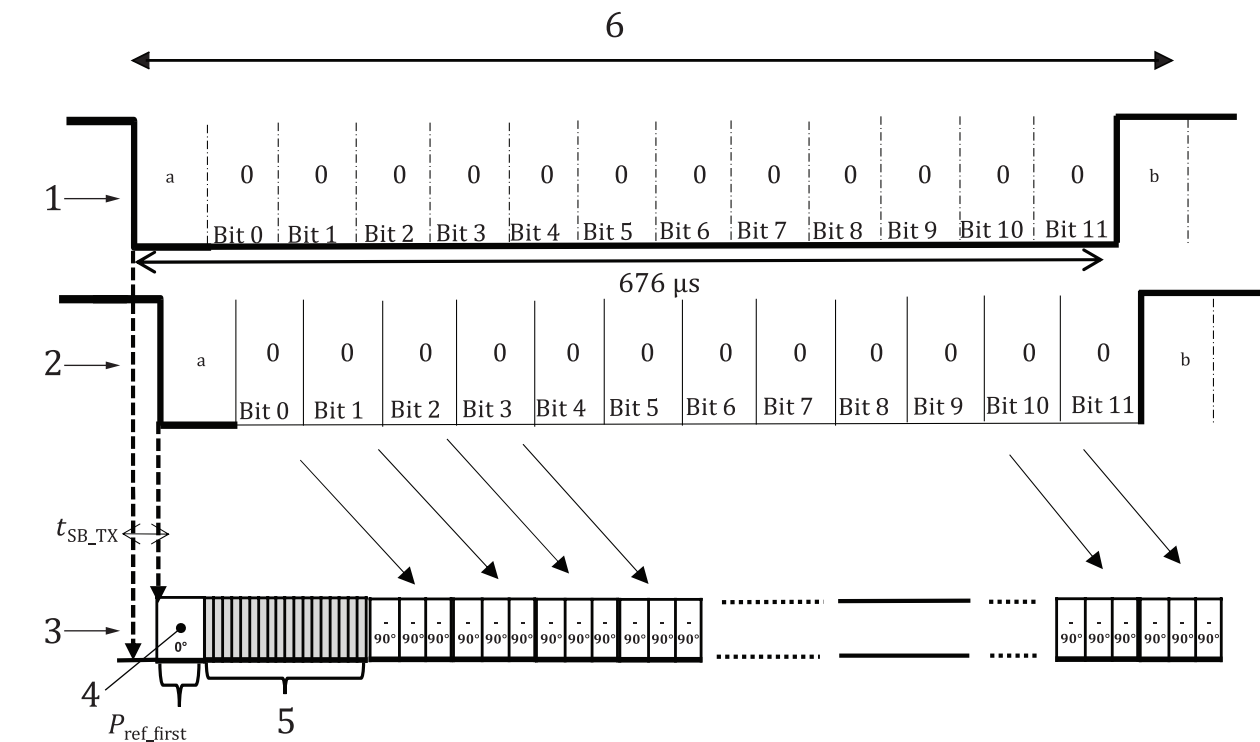
Key

- 1 TXD wake-up signal DC-LIN node A
- 2 RXD wake-up signal DC-LIN node A
- 3 DC powerline
- 4 ref phase 0°
- 5 byte field sync preamble
- 6 DC-LIN wake-up signal transmission of node A
- a Start.
- b Stop.

Figure 11 — Example of a wake-up signal modulation similar to byte field modulation

EXAMPLE [Figure 12](#) depicts a wake-up signal with the length of 676 μs. The wake-up signal is encoded by the DC-LIN EPL to a start bit, twelve '0' data bits and a stop bit (i.e. a break field).

The wake-up signal modulation is then similar to a break field modulation.



- Key**
- 1 TXD wake-up signal DC-LIN node A
 - 2 RXD wake-up signal DC-LIN node A
 - 3 DC powerline
 - 4 ref phase 0°
 - 5 byte field sync preamble
 - 6 DC-LIN wake-up signal transmission of node A
 - a Start.
 - b Stop.

Figure 12 — Example of a wake-up signal modulation similar to break field modulation

4.3 Timing requirements

4.3.1 Bit rate tolerance

The bit rate tolerance describes the allowed deviation of the actual bit rate from the nominal bit rate for a DC-LIN EPL.

Table 6 specifies the bit rate tolerances for both master node and slave node DC-LIN EPL.

Table 6 — DC-LIN EPL bit rate tolerances

Bit rate	Deviation
BR_9_6K	<±0,5 %
BR_10K	<±0,5 %
BR_19_2K	<±0,5 %

The specified DC-LIN EPL bit rate deviation provides sufficient nominal bit rate, which is complying with EPL sync byte field quantization automatic bit rate detection as specified in ISO 17987-4:2016, 5.2.2, and with EPL bit rate deviation specified in ISO 17987-4:2016, 5.1.

4.3.2 Bit timing

The bit time is defined according to the selected bit rate (given in [Table 6](#)).

Calculation of a bit time is given in [Formula \(5\)](#).

Definition of the formula:

$$t_{\text{BIT}} = 1/(\text{BR}) \quad (5)$$

4.3.3 Bit sample timing

The bit sample timing on TXD shall apply the bit sampling timing specified in ISO 17987-4:2016, 5.2.3.

4.3.4 TXD assert timeout event

The minimum TXD assert ('0') time without timeout event is given in [Formula \(6\)](#).

Definition of the formula:

$$t_{\text{txd_min_assert}} = 6 \text{ ms} \quad (6)$$

The minimum TXD deassert ('1') time after timeout event (i.e. timeout recovery time) is given in [Formula \(7\)](#).

Definition of the formula:

$$t_{\text{txd_min_recover}} = 10 t_{\text{BIT}} \quad (7)$$

Optionally, the DC-LIN EPL may detect a TXD assert timeout event when TXD assert time $> t_{\text{txd_min_assert}}$.

At the occurrence of TXD assert timeout event, the DC-LIN EPL's $T_{\text{tx_mode}}$ may be deactivated until the next falling edge on TXD, which occurs after TXD deassert ('1') time $> t_{\text{txd_min_recover}}$.

4.3.5 Delay between transmitted byte field (DC-LIN node A) and received byte field (DC-LIN node A)

The delay between transmitted byte field on TXD and the received byte field on RXD at a DC-LIN EPL node (locally) ($t_{\text{RX_delay_local}}$) is given in [Formula \(8\)](#).

Definition of the formula:

$$t_{\text{RX_delay_local}} = t_{\text{SB_TX}} \quad (8)$$

4.3.6 Delay between transmitted byte field (DC-LIN node A) and received byte field (DC-LIN node B)

The delay between the transmitted byte field, over to the DC powerline, and the receiver reconstructed byte field ($t_{\text{RX_delay}}$) is given in [Table 7](#).

The maximum process time at RX DC-LIN EPL side is given in [Formula \(9\)](#).

Definition of the formula:

$$t_{\text{rx_proc_max}} = 6/9 t_{\text{BIT}} \quad (9)$$

The minimum process time at RX DC-LIN EPL side is given in [Formula \(10\)](#).

Definition of the formula:

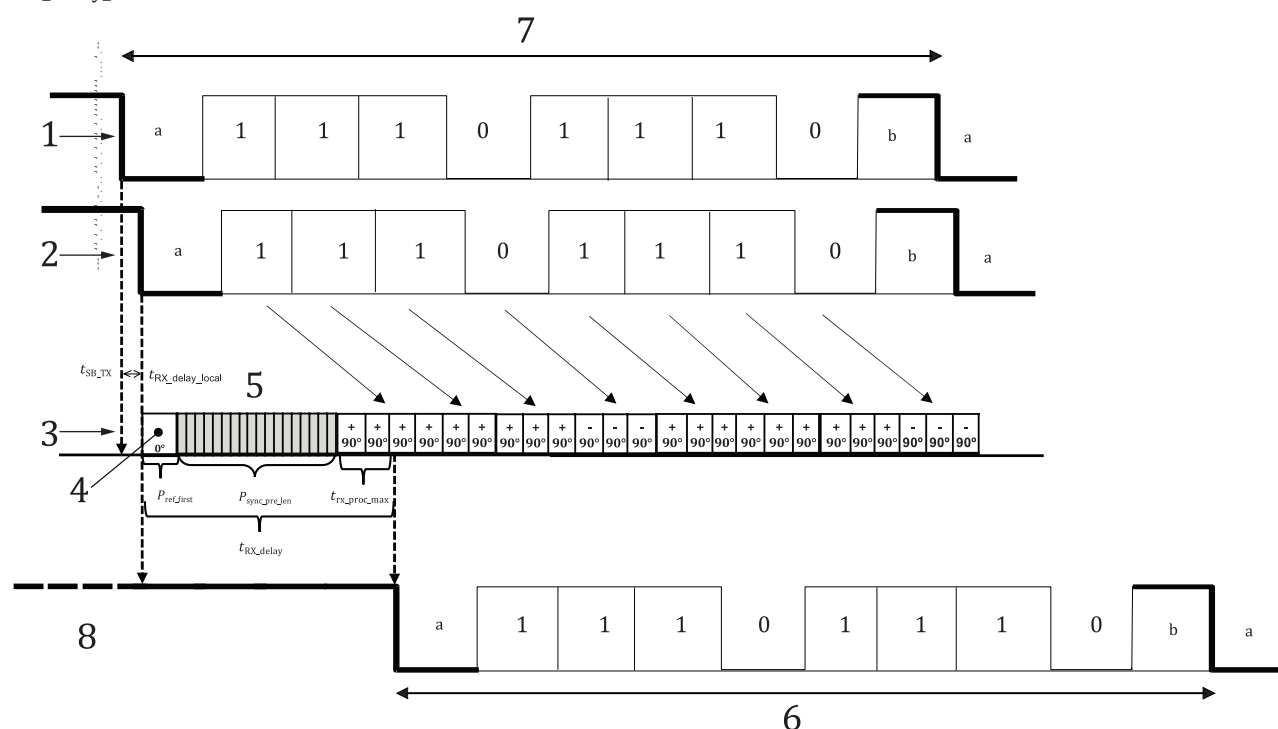
$$t_{RX_proc_min} = 2/9 t_{BIT} \quad (10)$$

(The transmission delay on the DC powerline is negligible as compared to t_{BIT} , and hence not included in t_{RX_delay} calculation).

Table 7 — Delay between TXD (DC-LIN EPL node A) to RXD (DC-LIN EPL node B)

Parameter	Min.	Typ.	Max.	Unit	Condition / comment
t_{RX_delay}	22/9	—	25/8	t_{BIT}	$t_{RX_delay_max} = t_{SB_TX} (2/16) + P_{ref_first} (3/9) + P_{sync_pre_len} (18/9) + t_{rx_proc_max} (6/9);$ $t_{RX_delay_min} = P_{ref_first} (2/9) + P_{sync_pre_len} (18/9) + t_{rx_proc_min} (2/9)$

Figure 13 shows an example of a single byte field transmission, demonstrating the $t_{RX_delay_max}$ and $t_{RX_delay_local}$.



Key

- 1 TXD DC-LIN node A
- 2 RXD DC-LIN node A
- 3 DC powerline
- 4 ref phase 0°
- 5 byte field sync preamble
- 6 DC-LIN byte field reception of node B
- 7 DC-LIN byte field transmission of node A
- 8 RXD DC-LIN node B
- a Start.
- b Stop.

Figure 13 — Delay between TXD and RXD over the powerline between a transmitting node A and receiving node B

4.4 Receiver characteristics

At the receiver node side, the DC-LIN EPL (e.g. node B) shall decode a received byte/break field modulation from DC-LIN EPL (e.g. node A) into bits transferred on RXD to the data link layer (see [Figures 8](#) and [9](#)).

DC-LIN EPL RX node (e.g. node B) error condition (RX_{err_cond}) is defined active when decoding of the received modulation from DC-LIN EPL (e.g. node A) results in detection of no phase shifts for the duration of at least $1 t_{BIT}$. RX_{err_cond} shall remain active for the remaining duration until completion of transmission $B_{fe_stop_bit}$ to the data link layer (see [Table 8](#) and [Figure 18](#)).

[Table 8](#) specifies RXD behaviour at receiver DC-LIN EPL node side.

If not otherwise stated, RXD logic level shall be set to '1'.

Table 8 — RXD behaviour at DC-LIN EPL receiver node side

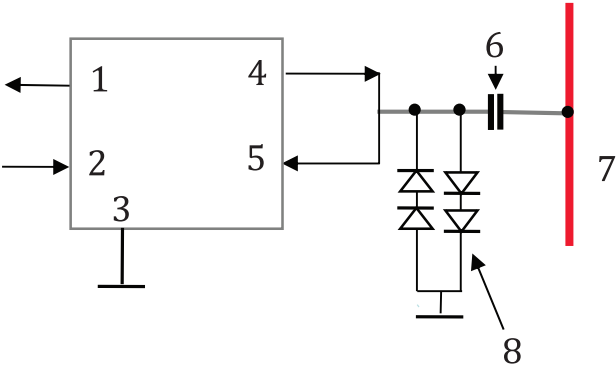
Parameter	RXD Logic level	Start time	Nominal bit length	Condition/comment
B_{start_bit}	Low ('0')	t_{RX_delay}	$1 t_{BIT}$	R_{rx_mode} is active
B_{data_bit}	Low ('0') when decoding bit modulation of data bit '0' or High ('1') when decoding bit modulation of data bit '1' or when RX_{err_cond} is active	$t_{RX_delay} + 1 t_{BIT}$	$1 t_{BIT}$	R_{rx_mode} is active The start time is defined for the first data bit (least significant bit), the rest of data bits are transmitted sequentially.
B_{stop_bit}	High ('1')	$t_{RX_delay} + 9 t_{BIT}$	$1 t_{BIT}$	R_{rx_mode} is active RX_{err_cond} is inactive Byte field receiving
$B_{fe_stop_bit}$	Low ('0')	$t_{RX_delay} + 9 t_{BIT}$	$1 t_{BIT}$	R_{rx_mode} is active RX_{err_cond} is active Byte field receiving
B_{del}	High ('1')	$t_{RX_delay} + 1 t_{BIT} + t_{break_field}$	$1 t_{BIT}$	R_{rx_mode} is active RX_{err_cond} is inactive Break field receiving

4.5 Electrical parameters

4.5.1 General configuration — Coupling to the DC powerline

The coupling to the DC powerline may include a coupling capacitor and a protection network.

[Figure 14](#) shows an example of a protection network using diodes.



- Key**
- 1 RXD receive data pin
 - 2 TXD transmit data pin
 - 3 GND
 - 4 line-out
 - 5 line-in
 - 6 coupling capacitor
 - 7 DC powerline
 - 8 protection network

Figure 14 — Example of powerline coupling circuit with protection network

4.5.2 Signal specification

Table 9 specifies the electrical parameters of the DC-LIN EPL signal on the line-out and line-in.

Table 9 — Electrical parameters of the DC-LIN EPL

Symbol	Description	Min.	Typ.	Max.	Unit	Condition/comment
Line-in	Input voltage level at line-in pin	0,005	—	1,44	V	Volt peak-to-peak Measurement-point is between line-in and the coupling capacitor with 50-Ω load (shown in Figure 15).
	Line-in input impedance	5	—	—	kΩ	
Line-out	Output voltage level at line-out pin	0,96	1,2	1,44	V	Volt peak-to-peak Measurement-point is between line-in and the coupling capacitor with 50-Ω load (shown in Figure 16). T_{tx_mode} is active
	Output voltage level at line-out pin	0	—	0,1	V	Volt peak-to-peak Measurement-point is between line-in and the coupling capacitor with 50-Ω load (shown in Figure 16). T_{tx_mode} is inactive or low-power mode is active
	Line-out input impedance	100	—	—	kΩ	T_{tx_mode} is inactive or low-power mode is active

Figure 15 illustrates DC-LIN EPL line-in measure point test set-up.

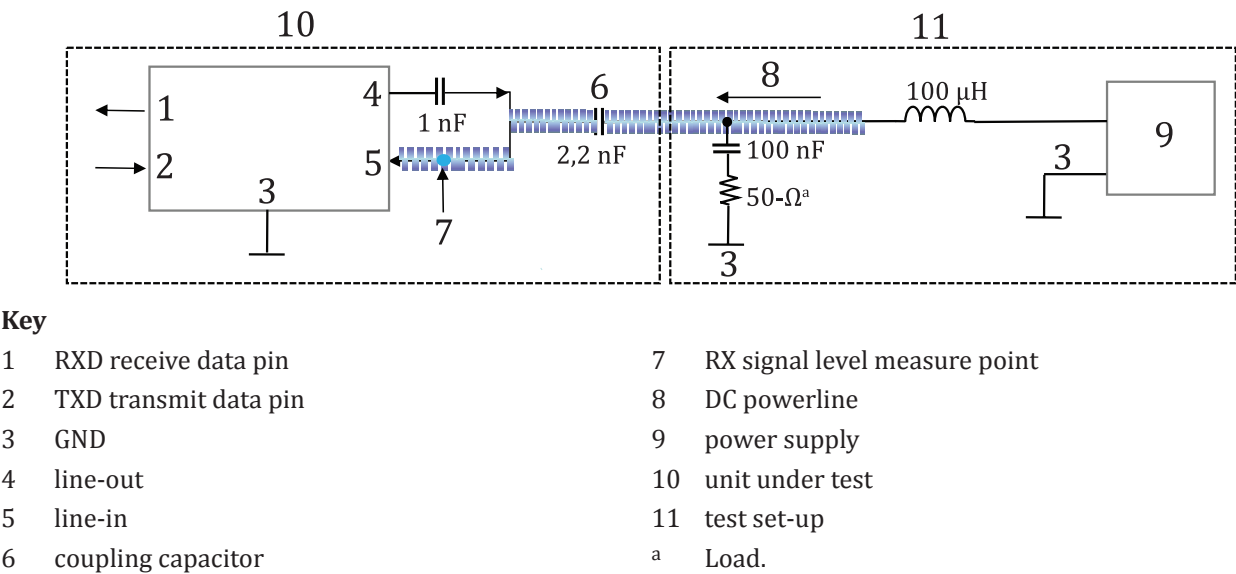


Figure 15 — DC-LIN EPL line-in measurement-point: test set-up

Figure 16 illustrates DC-LIN EPL line-out measure point test set-up.

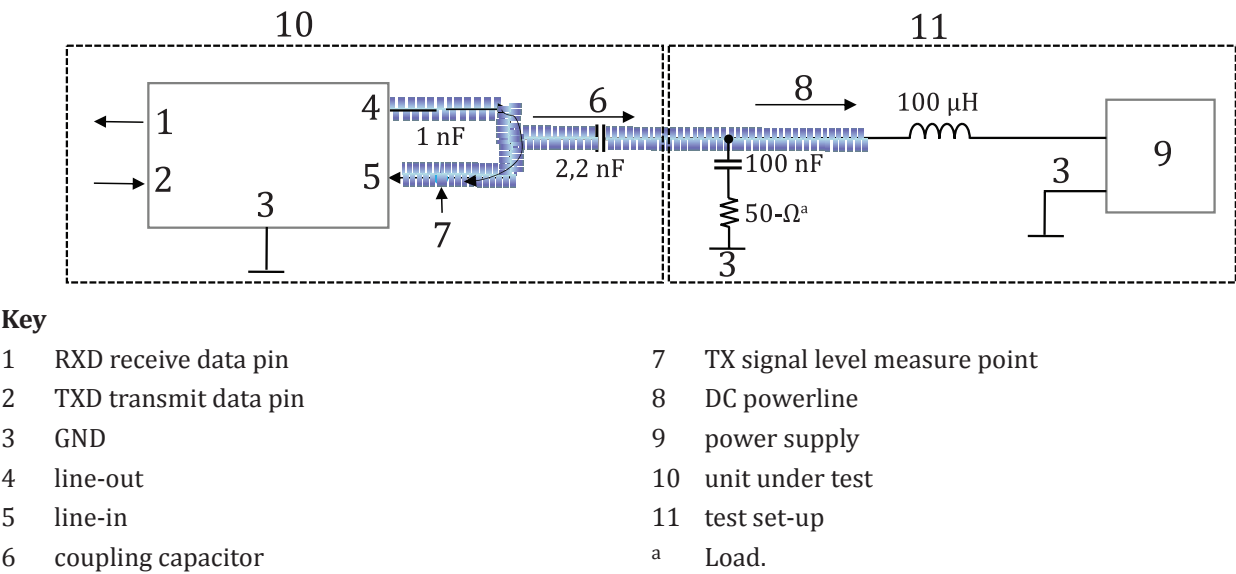


Figure 16 — DC-LIN EPL line-out measurement-point: test set-up

4.5.3 Line-out monitoring

The DC-LIN EPL shall monitor its transmitted output level at line-out. The DC-LIN EPL line-out monitoring function shall provide an error indication on RXD to the data link layer in case the sampled line-out peak level is below $L_{out_thr_lev}$.

Table 10 specifies the line-out monitoring peak threshold level ($L_{out_thr_lev}$).

Table 10 — Line-out monitoring peak threshold level

Parameter	Description	Min.	Typ.	Max.	Unit
$L_{\text{out_thr_lev}}$	Line-out monitoring peak threshold level	25	—	50	mV

Start time of line-out monitoring ($t_{\text{l_out_mon_start}}$) is given in [Formula \(11\)](#).

Definition of the formula:

$$t_{\text{l_out_mon_start}} = t_{\text{SB_TX}} \quad (11)$$

$t_{\text{SB_TX}}$ is given in [Table 3](#).

Line-out monitoring active duration ($t_{\text{l_out_mon_l}}$) from $t_{\text{SB_TX}}$ is given in [Formula \(12\)](#).

Definition of the formula:

$$t_{\text{l_out_mon_l}} = 10 t_{\text{BIT}} \quad (12)$$

The DC-LIN EPL shall sample the line-out level ($L_{\text{out_lev}}$) during line-out monitoring active duration at $t_{\text{l_out_mon_smp}}$ for each detected bit on TXD (see [Table 11](#)). The sampled monitoring level shall be kept until next $t_{\text{l_out_mon_smp}}$ (i.e. $t_{\text{l_out_mon_smp}} + t_{\text{BIT}}$).

[Table 11](#) specifies line-out monitoring sample time ($t_{\text{l_out_mon_smp}}$) with respect to $t_{\text{l_out_mon_start}}$.

Table 11 — Line-out monitoring sample time

Parameter	Description	Min.	Typ.	Max.	Unit
$t_{\text{l_out_mon_smp}}$	Line-out monitoring sample time	3/16	—	4/16	t_{BIT}

Line-out monitoring error condition ($L_{\text{out_err_cond}}$) is defined active when $L_{\text{out_lev}} < L_{\text{out_thr_lev}}$ and shall stay active until completion of $t_{\text{l_out_mon_l}}$. In case the transmit signal is constructed from more than one carrier frequency [i.e. $n > 1$, see [Formula \(1\)](#)], the $L_{\text{out_err_cond}}$ shall not be activated if at least one carrier frequency (e.g. c_2) out level is $\geq L_{\text{out_thr_lev}}$.

While $L_{\text{out_err_cond}}$ is active, and only after completion of byte field sync preamble transmission, the DC-LIN EPL shall change transmission from byte field modulation scheme to a constant phase (i.e. no phase shifts) transmission for the remaining transmission time (shown in [Figure 18](#)).

[Table 12](#) specifies RXD behaviour at transmitter DC-LIN EPL node while TXD is active and according to $L_{\text{out_lev}}$.

If not otherwise stated, RXD logic level shall be set to '1'.

Table 12 — RXD behaviour at DC-LIN EPL transmitter node side

Parameter	RXD Logic level	Start time	Nominal bit length	Condition/comment
$B_{\text{start_bit_local}}$	Low ('0')	$t_{\text{RX_delay_local}}$	$1 t_{\text{BIT}}$	$T_{\text{tx_mode}}$ is active $L_{\text{out_err_cond}}$ is inactive ($L_{\text{out_lev}} \geq L_{\text{out_thr_lev}}$)

Table 12 (continued)

Parameter	RXD Logic level	Start time	Nominal bit length	Condition/comment
B _{fe_start_bit_local}	High ('1')	$t_{l_out_mon_samp}$	1 t_{BIT}	T_{tx_mode} is active L_{out_err_cond} is active ($L_{out_lev} < L_{out_thr_lev}$) RXD indicates a start bit frame error to the data link layer.
B _{data_bit_local}	TXD	$t_{RX_delay_local} + 1 t_{BIT}$	1 t_{BIT}	T_{tx_mode} is active L_{out_err_cond} is inactive ($L_{out_lev} \geq L_{out_thr_lev}$) The start time is defined with respect to the first data bit (least significant bit), the other data bits are transmitted respectively with nominal bit length of 1 t_{BIT}.
B _{err_data_bit_local}	— TXD	$t_{l_out_mon_samp} + 1 t_{BIT}$	1 t_{BIT}	T_{tx_mode} is active L_{out_err_cond} is active ($L_{out_lev} < L_{out_thr_lev}$) The start time is defined with respect to the first data bit (least significant bit), the other data bits are transmitted respectively with nominal bit length of 1 t_{BIT}. RXD indicates an error to the data link layer.
B _{stop_bit_local}	High ('1')	$t_{RX_delay_local} + 9 t_{BIT}$	1 t_{BIT}	T_{tx_mode} is active L_{out_err_cond} is inactive ($L_{out_lev} \geq L_{out_thr_lev}$) Byte field transmission
B _{fe_stop_bit_local}	Low ('0')	$t_{l_out_mon_samp} + 9 t_{BIT}$	1 t_{BIT}	T_{tx_mode} is active L_{out_err_cond} is active ($L_{out_lev} < L_{out_thr_lev}$) The start time is defined with respect to $t_{l_out_mon_samp}$ of start bit. Byte field transmission RXD indicates a stop bit frame error to the data link layer.
B _{del_local}	High ('1')	$t_{RX_delay_local} + 1 t_{BIT} + t_{break_field}$	1 t_{BIT}	T_{tx_mode} is active L_{out_err_cond} is inactive ($L_{out_lev} \geq L_{out_thr_lev}$) Break field transmission

Table 12 (continued)

Parameter	RXD Logic level	Start time	Nominal bit length	Condition/comment
$B_{\text{fe-del_local}}$	Low ('0')	$t_{\text{l_out_mon_samp}} + 1 t_{\text{BIT}} + t_{\text{break_field}}$	$1 t_{\text{BIT}}$	$T_{\text{tx_mode}}$ is active $L_{\text{out_err_cond}}$ is active $(L_{\text{out_lev}} < L_{\text{out_thr_lev}})$ The start time is defined with respect to $t_{\text{l_out_mon_samp}}$ of start bit. Break field transmission RXD indicates a break field delimiter frame error to the data link layer.

Figure 17 shows an example of line-out monitoring; where line-out level is not less than $L_{\text{out_thr_lev}}$ ($L_{\text{out_err_cond}}$ is inactive).

The data link layer of node A transmits a byte field on TXD. Node A DC-LIN EPL line-out monitoring is activated from $t_{\text{SB_TX}}$ for the duration of $t_{\text{L_out_mon_l}}$, where the line-out level ($L_{\text{out_lev}}$) is being sampled at $t_{\text{l_out_mon_samp}}$ periodically every t_{BIT} .

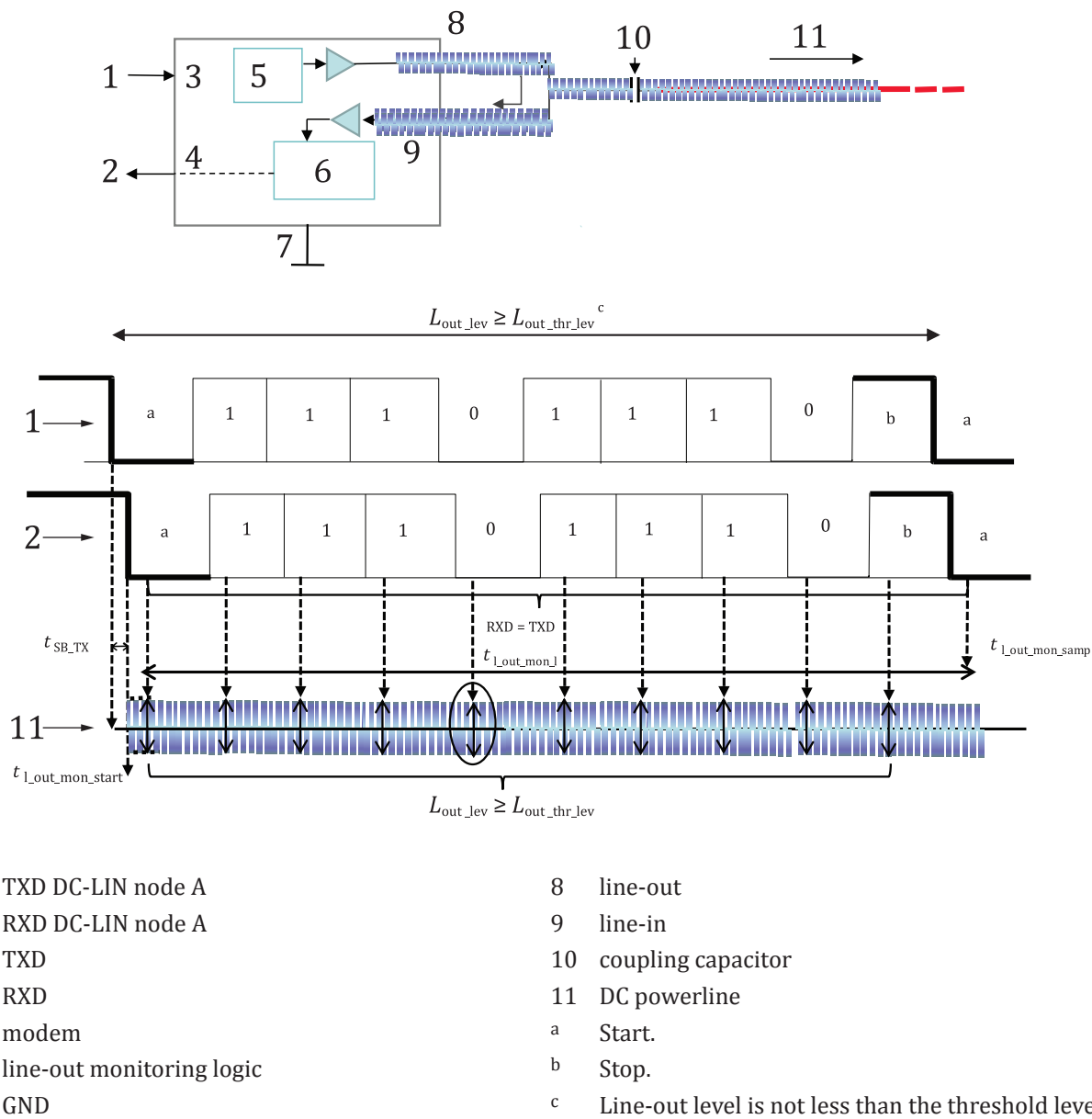
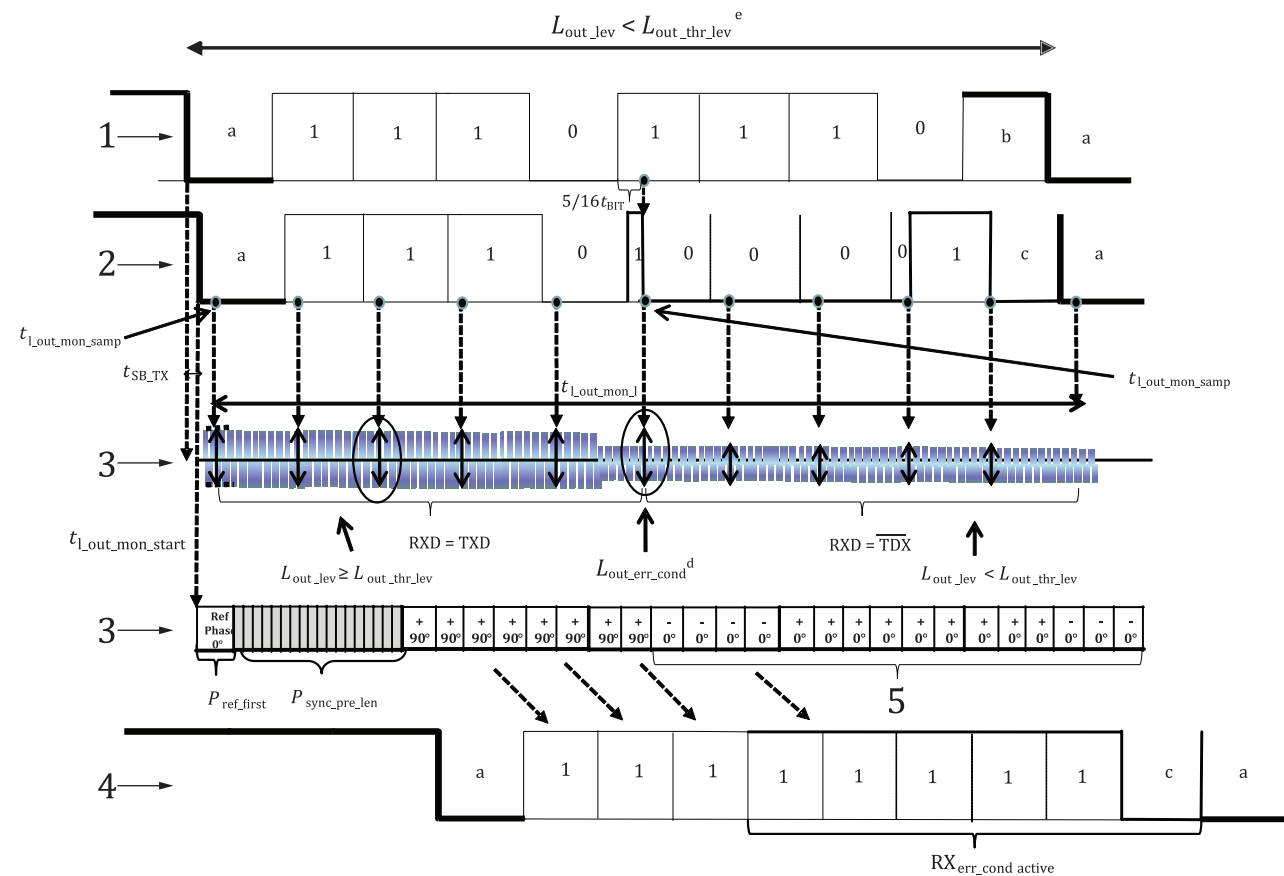


Figure 17 — Example of line-out monitoring: line-out level is not less than $L_{out_thr_lev}$

Figure 18 shows an example of line-out monitoring; where line-out level is lower than $L_{out_thr_lev}$ ($L_{out_err_cond}$ is active).



- Key**
- 1 TXD DC-LIN node A
 - 2 RXD DC-LIN node A
 - 3 DC powerline
 - 4 RXD DC-LIN node B
 - 5 no phase shifts
 - a Start.
 - b Stop.
 - c Stop bit frame error.
 - d Is triggered active.
 - e Line-out level is less than the threshold level.

Figure 18 — Example of line-out monitoring: line-out level is lower than $L_{out_thr_lev}$

4.5.4 ESD compliance

The DC-LIN EPL shall comply with requirements for protection against human body discharge according to IEC 61000-4-2. The minimum discharge voltage level shall be ± 2 kV.

4.6 Communication in the presence of faults

Table 13 specifies the DC-LIN EPL behaviour in the presence of faults, in which there shall be no damage to the DC-LIN EPL.

Table 13 — DC-LIN EPL behaviour in the presence of faults

Description of fault	Communication behaviour
DC powerline becomes shorted to ground.	Data communication is not possible.
DC powerline is stuck in asserted state.	Data communication is not possible.
DC-LIN EPL's TXD is continuously asserted.	Data communication is not possible unless affected node has implemented TXD asserted timeout. functionality (see 4.3.4).
DC powerline is disconnected from power.	Remaining powered DC-LIN EPLs continue to communicate with no degradation.

Annex A (informative)

DC-LIN EPL peripheral interface design considerations

An example for an interface between the DC-LIN EPL nodes and the DC powerline is shown in [Figure A.1](#).

The example shows three DC-LIN EPL nodes connected to the DC powerline. To reduce the attenuation of the DC-LIN EPL carrier signal by filtering capacitors in power supplies, inductors or low-frequency ferrite beads are added in series to the power supply.

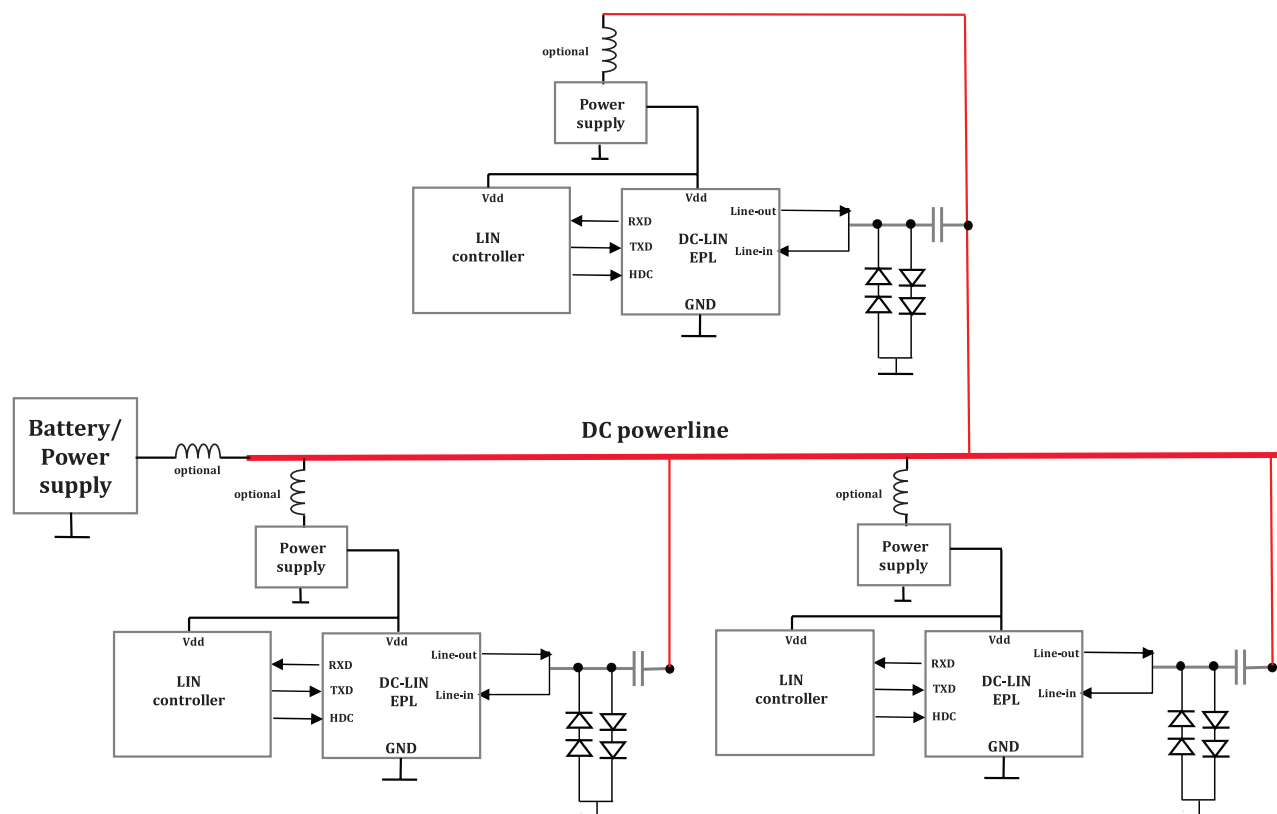


Figure A.1 — DC-LIN EPL coupling example

Annex B

(normative)

DC-LIN EPL conformance test plan

B.1 General

This annex specifies the DC-LIN EPL conformance test plan.

B.2 General test plan specification considerations

B.2.1 Test conditions

The tests shall be done at room temperature in a range of 15 °C to 35 °C.

B.2.2 Test case structure

In the description of each test case, it is specified, for which DC-LIN EPL type the test case is applicable, for master or slave node.

Each specification of a test case consists of nine parts.

- CTC # – Title
 - A common prefix for the conformance test case.
- Purpose
 - Brief statement outlining what the test case attempts to achieve.
- Reference
 - Defines the reference to this or other parts of the ISO 17987 series.
- Prerequisite
 - Defines the test set-up.
- Set-up
 - Defines the IUT configuration (see [B.2.4.5](#)).
- Step
 - Defines the test step/s for carrying out the test case.

- If more than one step is defined in this field, the steps shall be executed in the order as they are stated in the document.
- Iteration
 - Defines the number of repetitions of test procedure test steps.
- Expected response
 - Defines the expected behaviour of the IUT when executing the test steps.
- Remark
 - Contains a description of known issues with the test case, which can affect test results in certain situations.

B.2.3 IUT general specification

IUT with RXD and TXD connectors available and line-in and line-out connector available is applicable for the DC-LIN EPL test cases specified in this document.

B.2.4 Test system requirements

B.2.4.1 Generation of LIN frames

The test system shall ensure the precision of the bit time of a master node as specified in this document.

B.2.4.2 Normal requirements for the test cases

For proper measurement and verification of tests pattern (see [Table B.1](#)), the test system shall use a minimum over sampling factor of 16; see [Formula \(B.1\)](#).

The definition of the sample resolution is given in [Formula \(B.1\)](#).

Definition of the formula:

$$\text{sample resolution} = t_{\text{BIT}} / 16 \quad (\text{B.1})$$

B.2.4.3 Special requirements for bit timing testing

For proper measurement and verification of the bit timing tests, the test system shall measure with a minimum over sampling of 10 to the precision of a DC-LIN EPL bit rate deviation given in this document, [Table 6](#); see [Formula \(B.2\)](#).

The definition of the bit time sample resolution is given in [Formula \(B.2\)](#).

Definition of the formula:

$$\text{bit time sample resolution} = t_{\text{BIT}} \times 0,000\ 5 \quad (\text{B.2})$$

B.2.4.4 Test system configuration

A detailed test system configuration is given under the test cases definitions. The test system shall be implemented in a way that the test cases can be performed as specified and reproducible.

B.2.4.5 General predefinitions for IUT configuration

B.2.4.5.1 General

Before executing a test case, some settings shall be done by LT concerning:

- the IUT initialization:
 - an initialization of the IUT shall be performed by the LT before each test case. To initialize the IUT a reset is carried out and thereafter the IUT shall be reconfigured;
- the IUT as master or slave node:
 - specifies to what state the IUT shall have been set before starting the execution of the test. If not otherwise defined, the IUT as master node T_{tx_mode} is active and the IUT as slave node R_{rx_mode} is active;
- the test pattern to be sent by the LT on TXD of the IUT or to stimulate the IUT line-in;
- the IUT bit rate configuration;
- the IUT transmit signal number of carrier frequencies;
- the IUT working carrier frequency configuration.

B.2.4.5.2 IUT TXD test pattern specification

[Table B.1](#) specifies the test pattern to be sent on TXD of IUT in test cases.

Table B.1 — IUT TXD test pattern specification

Test pattern	Requirements for the test pattern
TST_BYTE_55	Single byte field 55 ₁₆
TST_BYTE_7A	Single byte field 7A ₁₆
TST_BYTE_65	Single byte field 65 ₁₆
TST_BYTE_5F	Single byte field 5F ₁₆
TST_BREAK_FIELD	Break field with length of 13 t_{BIT} and break delimiter of 1 t_{BIT}
TST_WAKEUP_SIGNAL_SHORT	Wake-up signal with length of 260 μ s
TST_WAKEUP_SIGNAL_LONG	Wake-up signal with length of 4 992 μ s

B.2.4.5.3 Transmit signal test pattern specification

The transmit signal test pattern shall comply with transmitter characteristics specified in this document (see [4.2](#), and its subclauses).

[Table B.2](#) specifies the transmit signal test pattern to be sent over the powerline in test cases.

Table B.2 — Transmit signal test pattern specification

Test pattern	Requirements for the transmit signal test pattern	Transmit signal phases	Phase length	Remarks
TXS_S_BYTE_55	Single byte field modulation of byte field 55 ₁₆	0°	1/3 t_{BIT}	$P_{\text{ref_first}}$
		90°, 180°, 90°, 180°, 90°, 0°, 90°, 180°, 90°, 180°, 90°, 0°, 90°, 180°, 90°, 180°, 90°, and 0°.	1/9 t_{BIT}	$P_{\text{sync_p}}$
		90°, 180°, 270°, 180°, 90°, 0°, 90°, 180°, 270°, 180°, 90°, 0°, 90°, 180°, 270°, 180°, 90°, 0°, 90°, 180°, 270°, 180°, 90°, and 0°.	1/3 t_{BIT}	P_{data}
TXS_BYTE_7A_CONSEC_55	Two consecutive bytes field modulation. First byte field 7A ₁₆ with an inter-byte space no longer than 1/3 t_{BIT} , followed by byte field 55 ₁₆ .	0°	1/3 t_{BIT}	$P_{\text{ref_first}}$ of the first byte field
		90°, 180°, 90°, 180°, 90°, 0°, 90°, 180°, 90°, 180°, 90°, 0°, 90°, 180°, 90°, 180°, 90°, and 0°.	1/9 t_{BIT}	$P_{\text{sync_p}}$ of the first byte field
		90°, 180°, 270°, 0°, 90°, 180°, 270°, 0°, 90°, 0°, 270°, 180°, 90°, 0°, 270°, 0°, 90°, 180°, 90°, 0°, 270°, 0°, 90°, and 180° ($P_{\text{ref_consec}}$).	1/3 t_{BIT}	P_{data} of the first byte field Last transmitted P_{data} is the $P_{\text{ref_consec}}$ of the second byte field.
		270°, 0°, 270°, 0°, 270°, 180°, 270°, 0°, 270°, 0°, 270°, 180°, 270°, 0°, 270°, 0°, 270°, and 180°.	1/9 t_{BIT}	$P_{\text{sync_p}}$ of the second byte field
		270°, 0°, 90°, 0°, 270°, 180°, 270°, 0°, 90°, 0°, 270°, 180°, 270°, 0°, 90°, 0°, 270°, 180°, 270°, 0°, 90°, 0°, 270°, and 180°.	1/3 t_{BIT}	P_{data} of the second byte field

Table B.2 (continued)

Test pattern	Requirements for the transmit signal test pattern	Transmit signal phases	Phase length	Remarks
TXS_BYTE_55_CONSEC_55	Three consecutive bytes field modulation of byte field 55 ₁₆ with an inter-byte space no longer than $1/3 t_{\text{BIT}}$ between the first byte and the second byte, and with an inter-byte space longer than $1/3 t_{\text{BIT}}$ between the second byte and the third byte	0°	$1/3 t_{\text{BIT}}$	$P_{\text{ref_first}}$ of the first byte field
		90°, 180°, 90°, 180°, 90°, 0°, 90°, 180°, 90°, 180°, 90°, 0°, 90°, 180°, 90°, 180°, 90°, and 0°.	$1/9 t_{\text{BIT}}$	$P_{\text{sync_p}}$ of the first byte field
		90°, 180°, 270°, 180°, 90°, 0°, 90°, 180°, 270°, 180°, 90°, 0°, 90°, 180°, 270°, 180°, 90°, and 0° ($P_{\text{ref_consec}}$).	$1/3 t_{\text{BIT}}$	P_{data} of the first byte field Last transmitted P_{data} is the $P_{\text{ref_consec}}$ of the second byte field.
		90°, 180°, 90°, 180°, 90°, 0°, 90°, 180°, 90°, 180°, 90°, 0°, 90°, 180°, 90°, 180°, 90°, and 0°.	$1/9 t_{\text{BIT}}$	$P_{\text{sync_p}}$ of the second byte field
		90°, 180°, 270°, 180°, 90°, 0°, 90°, 180°, 270°, 180°, 90°, 0°, 90°, 180°, 270°, 180°, 90°, and 0°.	$1/3 t_{\text{BIT}}$	P_{data} of the second byte field
		0°	$1/3 t_{\text{BIT}}$	$P_{\text{ref_first}}$ of the third byte field
		90°, 180°, 90°, 180°, 90°, 0°, 90°, 180°, 90°, 180°, 90°, 0°, 90°, 180°, 90°, 180°, 90°, and 0°.	$1/9 t_{\text{BIT}}$	$P_{\text{sync_p}}$ of the third byte field
		90°, 180°, 270°, 180°, 90°, 0°, 90°, 180°, 270°, 180°, 90°, 0°, 90°, 180°, 270°, 180°, 90°, and 0°.	$1/3 t_{\text{BIT}}$	P_{data} of the third byte field
TXS_S_BYTE_55_ERR_COND_N2	Single byte field modulation of byte field 55 ₁₆ with $L_{\text{out_err_cond}}$ triggered at data bit 2.	0°	$1/3 t_{\text{BIT}}$	$P_{\text{ref_first}}$
		90°, 180°, 90°, 180°, 90°, 0°, 90°, 180°, 90°, 180°, 90°, 0°, 90°, 180°, 90°, 180°, 90°, and 0°.	$1/9 t_{\text{BIT}}$	$P_{\text{sync_p}}$
		90°, 0°, 0°, 0°, 0°, 0°, 0°, 0°, 0°, 0°, 0°, 0°, 0°, 0°, 0°, 0°, 0°, 0°, and 0°.	$1/3 t_{\text{BIT}}$	P_{data}
TXS_S_BYTE_55_ERR_COND_N4	Single byte field modulation of byte field 55 ₁₆ with $L_{\text{out_err_cond}}$ triggered at data bit 4.	0°	$1/3 t_{\text{BIT}}$	$P_{\text{ref_first}}$
		90°, 180°, 90°, 180°, 90°, 0°, 90°, 180°, 90°, 180°, 90°, 0°, 90°, 180°, 90°, 180°, 90°, and 0°.	$1/9 t_{\text{BIT}}$	$P_{\text{sync_p}}$
		90°, 180°, 270°, 180°, 90°, 0°, 90°, 0°, 0°, 0°, 0°, 0°, 0°, 0°, 0°, 0°, 0°, 0°, 0°, and 0°.	$1/3 t_{\text{BIT}}$	P_{data}
TXS_S_BYTE_55_ERR_COND_N6	Single byte field modulation of byte field 55 ₁₆ with $L_{\text{out_err_cond}}$ triggered at data bit 6.	0°	$1/3 t_{\text{BIT}}$	$P_{\text{ref_first}}$
		90°, 180°, 90°, 180°, 90°, 0°, 90°, 180°, 90°, 180°, 90°, 0°, 90°, 180°, 90°, 180°, 90°, and 0°.	$1/9 t_{\text{BIT}}$	$P_{\text{sync_p}}$
		90°, 180°, 270°, 180°, 90°, 0°, 90°, 180°, 270°, 180°, 90°, 0°, 90°, 0°, 0°, 0°, 0°, 0°, 0°, 0°, and 0°.	$1/3 t_{\text{BIT}}$	P_{data}

Table B.2 (continued)

Test pattern	Requirements for the transmit signal test pattern	Transmit signal phases	Phase length	Remarks
TXS_S_BYTE_7A	Single byte field modulation of byte field 7A ₁₆	0°	1/3 t_{BIT}	$P_{\text{ref_first}}$
		90°, 180°, 90°, 180°, 90°, 0°, 90°, 180°, 90°, 180°, 90°, 0°, 90°, 180°, 90°, 180°, 90°, and 0°.	1/9 t_{BIT}	$P_{\text{sync_p}}$
		270°, 180°, 90°, 180°, 270°, 0°, 270°, 180°, 90°, 180°, 270°, 0°, 90°, 180°, 270°, 0°, 90°, 180°, 270°, and 180°	1/3 t_{BIT}	P_{data}
TXS_S_BYTE_65	Single byte field modulation of byte field 65 ₁₆	0°	1/3 t_{BIT}	$P_{\text{ref_first}}$
		90°, 180°, 90°, 180°, 90°, 0°, 90°, 180°, 90°, 180°, 90°, 0°, 90°, 180°, 90°, 180°, 90°, and 0°.	1/9 t_{BIT}	$P_{\text{sync_p}}$
		90°, 180°, 270°, 180°, 90°, 0°, 90°, 180°, 270°, 180°, 90°, 0°, 270°, 180°, 90°, 180°, 270°, 0°, 90°, 180°, 270°, 180°, 90°, and 0°.	1/3 t_{BIT}	P_{data}
TXS_S_BYTE_5F	Single byte field modulation of byte field 5F ₁₆	0°	1/3 t_{BIT}	$P_{\text{ref_first}}$
		90°, 180°, 90°, 180°, 90°, 0°, 90°, 180°, 90°, 180°, 90°, 0°, 90°, 180°, 90°, 180°, 90°, and 0°.	1/9 t_{BIT}	$P_{\text{sync_p}}$
		90°, 180°, 270°, 0°, 90°, 180°, 270°, 0°, 90°, 180°, 270°, 0°, 90°, 180°, 270°, 180°, 90°, 0°, 90°, 180°, 270°, 180°, 90°, and 0°.	1/3 t_{BIT}	P_{data}
TXS_BREAK_FIELD	Break field modulation with length of 13 t_{BIT}	0°	1/3 t_{BIT}	$P_{\text{ref_first}}$
		90°, 180°, 90°, 180°, 90°, 0°, 90°, 180°, 90°, 180°, 90°, 0°, 90°, 180°, 90°, 180°, 90°, and 0°.	1/9 t_{BIT}	$P_{\text{sync_p}}$
		270°, 180°, 90°, 0°, 270°, 180°, 90°, 0°, 270°, 180°, 90°, 0°, 270°, 180°, 90°, 0°, 270°, 180°, 90°, 0°, 270°, 180°, 90°, and 0°.	1/3 t_{BIT}	P_{data} 12 data bit '0'
TXS_WAKEUP_SIGNAL_SHORT	Wake-up signal modulation with length of 260 μs	0°	1/3 t_{BIT}	$P_{\text{ref_first}}$
		90°, 180°, 90°, 180°, 90°, 0°, 90°, 180°, 90°, 180°, 90°, 0°, 90°, 180°, 90°, 180°, 90°, and 0°.	1/9 t_{BIT}	$P_{\text{sync_p}}$
		270°, 180°, 90°, 0°, 270°, 180°, 90°, 0°, 270°, 180°, 90°, 0°, 90°, 180°, 270°, 0°, 90°, 180°, 270°, and 0°.	1/3 t_{BIT}	P_{data} 4 data bit '0' 4 data bit '1'

Table B.2 (continued)

[illegible]

B.2.4.5.4 Default delays for frame headers

If not otherwise specified, the delay between frame headers is specified in [Table B.3](#).

Table B.3 — Default delays for frame headers

Bit rate	Delay
BR_19_2K	15 ms
BR_10K	20 ms
BR_9_6_K	20 ms

B.2.4.5.5 Default spaces between LIN frame fields

If not otherwise specified in the test cases, the inter-byte spaces between the header, the response field and the data fields are set to zero.

B.2.4.5.6 IUT default bit rate

The default bit rate is set to the highest bit rate that is specified by the specification of the IUT (i.e. BR_19_2K). Test cases with a different bit rate setting can only be performed when the required bit rate is in accordance with the specification of the IUT.

B.2.4.5.7 IUT default transmit signal number of carrier frequencies

If not specified otherwise in the test case, the default number of the IUT transmit signal carrier frequencies is set to 1.

$CF_{\max}$ is defined as the maximal number of carrier frequencies implemented in IUT transmit signal.

Test case specified with higher number of carrier frequencies setting can only be performed when the required number of carrier frequencies is in accordance with the specification and implementation of the IUT $CF_{\max}$.

B.2.4.5.8 IUT default transmit signal carrier frequency

[Table B.4](#) specifies the IUT default carrier frequencies configuration according to the IUT $CF_{\max}$ as specified in [Table 1](#).

Table B.4 — IUT Default carrier frequencies configuration

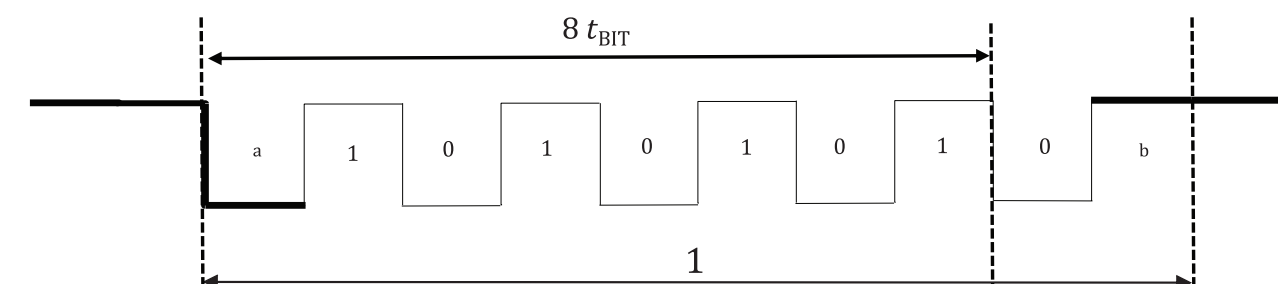
Number of transmit signal carriers ($CF_{\max}$)	Default carrier frequencies configuration	Unit
1	13	MHz
2	13 and 20	
3	13, 20 and 30	
4	5, 13, 20 and 30	

The maximal number of carrier frequencies ($CF_{\max}$) constructing a DC-LIN EPL transmit signal is subject to the IUT implementation.

B.2.4.5.9 DC-LIN EPL TXD and RXD time measurements requirements

The time measurements of IUT TXD and RXD as master or slave node shall be done on TST_BYTE_55, between the start bit falling edge and the 5th falling edge, which is available in a distance of eight bit times. This allows a simple calculation of the basic bit time t_{BIT} .

[Figure B.1](#) shows an example of TST_BYTE_55 byte field for time measurement.



Key

1 TST_BYTE_55

a Start.

b Stop.

Figure B.1 — Example of TST_BYTE_55 for time measurement

B.2.4.5.10 DC-LIN EPL line-out and line-in measurements requirements

[Table B.5](#) specifies the line-out and line-in measurements requirements.

Table B.5 — Line-out and line-in measurements - Requirements

Measurement type	Minimum requirement
Carrier frequency	Oscilloscope: — 250 Mega samples/s — Rise time $\leq 3,5$ ns — Bandwidth: 50 MHz — Voltage per division: 1 mV to 2 V
Phase length	
Phase shift value	

B.3 Transmitter characteristics timing

B.3.1 General

This subclause verifies if the DC-LIN EPL transmitter characteristics timing parameters of the IUT as master or slave node conform to the transmitter characteristics timing parameters that are specified in this document. The IUT is stimulated to transmit defined byte field/break field (lower tester) and the resulting behaviour is observed and compared with the expected behaviour (upper tester).

Figure B.2 depicts the test configuration of the test system 'Transmitter characteristics timing'.

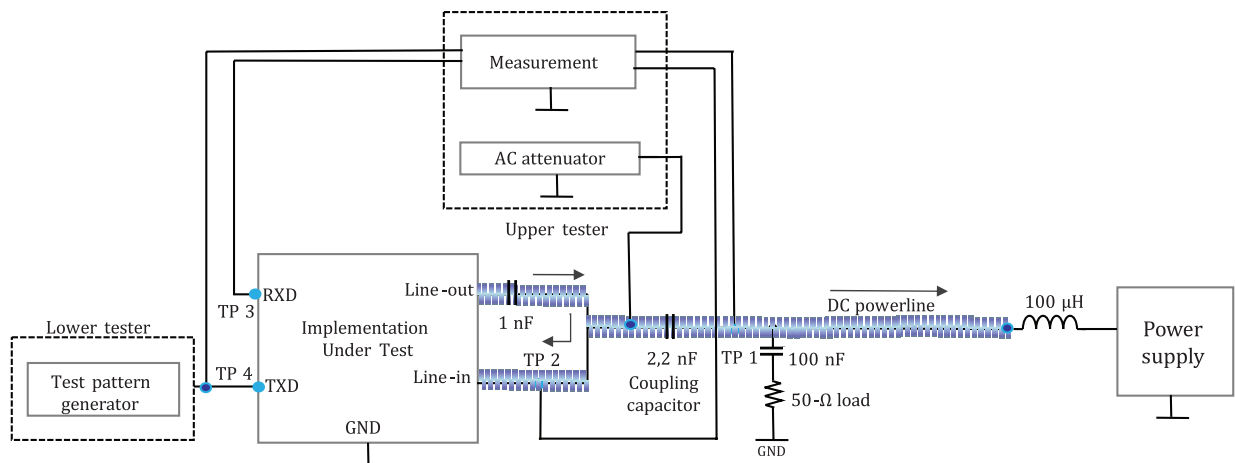


Figure B.2 — Test system: Transmitter characteristics timing

B.3.2 Transmit signal carrier frequencies

Table B.6 defines the test structure of CTC_TX_CHARC_1.

Table B.6 — Test structure of CTC_TX_CHARC_1

Item	Content
CTC # - Title	1.CTC_TX_CHARC_1 - Transmit signal carrier frequencies
Purpose	This CTC verifies the correct carrier frequencies transmission, constructing the DC-LIN EPL transmit signal.
Reference	4.2.1
Prerequisite	1 LT with the capability to stimulate, reset and configure the IUT. 2 The UT is able to provide time measurement capabilities synchronised with the test step events (see Figure B.2).
Set-up	1 LT: Shall configure the IUT with default bit rate (see B.2.4.5.5). 2 LT: Shall configure the IUT with default working carrier frequencies (see Table B.4). 3 LT: Shall configure the IUT with CF_{max} transmit signal number of carrier frequencies (i.e. 1, 2, 3, or 4).
Step	1 LT: Shall send repeated TST_BYTE_55 on IUT TXD. 2 UT: Shall measure the corresponding carrier frequencies at TP 1 (see Figure B.2).
Iteration	Not applicable
Expected response	Step 2 The corresponding carrier frequencies measured at TP 1 (see Figure B.2) complies with Table 1 = OK.
Remark	—

B.3.3 Start time of byte field/s transmission

Table B.7 defines the test structure of CTC_TX_CHARS_2.

Table B.7 — Test structure of CTC_TX_CHARS_2

Item	Content
CTC # – Title	1.CTC_TX_CHARS_2 – Start time of byte field/s transmission
Purpose	This CTC verifies that the start time of byte field/s transmission is according to t_{SB_TX} .
Reference	4.2.2.3 and 4.3.5
Prerequisite	1 LT with the capability to stimulate, reset and configure the IUT. 2 The UT is able to provide time measurement capabilities synchronised with the test step events (see Figure B.2).
Set-up	1 LT: Shall configure the IUT with default bit rate (see B.2.4.5.5). 2 LT: Shall configure the IUT with default working carrier frequencies (see Table B.4). 3 LT: Shall configure the IUT with CF_{max} transmit signal number of carrier frequencies (i.e. 1, 2, 3, or 4).
Step	1 LT: Shall send TST_BYTE_55 on IUT TXD. 2 UT: Shall measure the time from falling edge of start bit to IUT start of transmission at line-out at TP 1. 3 UT: Shall measure the time from falling edge of start bit to IUT falling edge of start bit at RXD at TP 3. 4 LT: Shall send two consecutive TST_BYTE_55 on IUT TXD with an inter-byte space no longer than $1/3 t_{BIT}$. 5 UT: Shall measure the time from falling edge of start bit to IUT start of transmission at line-out at TP 1. 6 UT: Shall measure the time from falling edge of start bit to IUT falling edge of start bit at RXD at TP 3. 7 LT: Shall send three consecutive TST_BYTE_55 on IUT TXD with an inter-byte space no longer than $1/3 t_{BIT}$ between the first byte and the second byte, and with an inter-byte space longer than $1/3 t_{BIT}$ between the second byte and the third byte. 8 UT: Shall measure the time from falling edge of start bit to IUT start of transmission at line-out at TP 1. 9 UT: Shall measure the time from falling edge of start bit to IUT falling edge of start bit at RXD at TP 3.
Iteration	Not applicable
Expected response	- Step 2 The time from falling edge of start bit to IUT start of transmission at line-out measured at TP 1 complies with $t_{SB_TX} = OK$. - Step 3 The time from falling edge of start bit to IUT falling edge of start bit at RXD measured at TP 3 complies with $t_{RX_delay_local} = OK$. - Step 5 The time from falling edge of start bit to IUT start of transmission at line-out measured at TP 1 complies with $t_{SB_TX} = OK$. - Step 6 The time from falling edge of start bit to IUT falling edge of start bit at RXD measured at TP 3 complies with $t_{RX_delay_local} = OK$. - Step 8 The time from falling edge of start bit to IUT start of transmission at line-out measured at TP 1 complies with $t_{SB_TX} = OK$. - Step 9 The time from falling edge of start bit to IUT falling edge of start bit at RXD measured at TP 3 complies with $t_{RX_delay_local} = OK$.
Remark	—

B.3.4 Length and value of reference phase

Table B.8 defines the test structure of CTC_TX_CHARS_3.

Table B.8 — Test structure of CTC_TX_CHARC_3

Item	Content
CTC # - Title	1.CTC_TX_CHARC_3 - Length and value of reference phase
Purpose	This CTC verifies the correct length and value of the reference phase.
Reference	4.2.2.2
Prerequisite	1 LT with the capability to stimulate, reset and configure the IUT. 2 The UT is able to provide time measurement capabilities synchronised with the test step events (see Figure B.2).
Set-up	1 LT: Shall configure the IUT with default bit rate (see B.2.4.5.5). 2 LT: Shall configure the IUT with default working carrier frequencies (see Table B.4). 3 LT: Shall configure the IUT with default transmit signal number of carrier frequencies.
Step	1 LT: Shall send TST_BYTE_55 on IUT TXD. 2 UT: Shall measure the length and value of reference phase at TP 1. 3 LT: Shall send TST_BYTE_7A on IUT TXD with an inter-byte space no longer than $1/3 t_{BIT}$ followed by TST_BYTE_55. 4 UT: Shall measure the length and value of first byte reference phase at TP 1. 5 UT: Shall measure the length and value of second byte reference phase at TP 1. 6 LT: Shall send three consecutive TST_BYTE_55 on IUT TXD with an inter-byte space no longer than $1/3 t_{BIT}$ between the first byte and the second byte, and with an inter-byte space longer than $1/3 t_{BIT}$ between the second byte and the third byte. 7 UT: Shall measure the length and value of first byte reference phase at TP 1. 8 UT: Shall measure the length and value of second byte reference phase at TP 1. 9 UT: Shall measure the length and value of third byte reference phase at TP 1.
Iteration	Not applicable
Expected response	- Step 2 The length of reference phase measured at TP 1 complies with $P_{ref_first} = OK$, and the value of reference phase is $0^\circ = OK$. - Step 4 The length of first byte reference phase measured at TP 1 complies with $P_{ref_first} = OK$, and the value of reference phase is $0^\circ = OK$. - Step 5 The length of second byte reference phase measured at TP 1 complies with $P_{ref_consec} = OK$, and the value of reference phase is $180^\circ = OK$. - Step 7 The length of first byte reference phase measured at TP 1 complies with $P_{ref_first} = OK$, and the value of reference phase is $0^\circ = OK$. - Step 8 The length of second byte reference phase measured at TP 1 complies with $P_{ref_consec} = OK$, and the value of reference phase is $0^\circ = OK$. - Step 9 The length of third byte reference phase measured at TP 1 complies with $P_{ref_first} = OK$, and the value of reference phase is $0^\circ = OK$.
Remark	—

B.3.5 Length of byte field sync preamble phases

[Table B.9](#) defines the test structure of CTC_TX_CHARC_4.

Table B.9 — Test structure of CTC_TX_CHARC_4

Item	Content
CTC # - Title	1.CTC_TX_CHARC_4 - Length of byte field sync preamble phases
Purpose	This CTC verifies the correct length of the byte field sync preamble phases.
Reference	Table 4

Table B.9 (continued)

Item	Content
CTC # – Title	1.CTC_TX_CHARC_4 – Length of byte field sync preamble phases
Prerequisite	1 LT with the capability to stimulate, reset and configure the IUT. 2 The UT is able to provide time measurement capabilities synchronised with the test step events (see Figure B.2).
Set-up	1 LT: Shall configure the IUT with default bit rate (see B.2.4.5.5). 2 LT: Shall configure the IUT with default working carrier frequencies (see Table B.4). 3 LT: Shall configure the IUT with default transmit signal number of carrier frequencies.
Step	1 LT: Shall send TST_BYTE_55 on IUT TXD. 2 UT: Shall measure the length of each byte field sync preamble phases at TP 1.
Iteration	Not applicable
Expected response	Step 2 The length of each byte field sync preamble phases measured at TP 1 complies with Table 4 with tolerance of $\pm 0,5\%$ = OK.
Remark	—

B.3.6 Values of byte field sync preamble phase shifts

[Table B.10](#) defines the test structure of CTC_TX_CHARC_5.

Table B.10 — Test structure of CTC_TX_CHARC_5

Item	Content
CTC # – Title	1.CTC_TX_CHARC_5 – Values of byte field sync preamble phase shifts
Purpose	This CTC verifies the correct values of the byte field sync preamble phase shifts.
Reference	Table 4
Prerequisite	1 LT with the capability to stimulate, reset and configure the IUT. 2 The UT is able to provide time measurement capabilities synchronised with the test step events (see Figure B.2).
Set-up	1 LT: Shall configure the IUT with default bit rate (see B.2.4.5.5). 2 LT: Shall configure the IUT with default working carrier frequencies (see Table B.4). 3 LT: Shall configure the IUT with default transmit signal number of carrier frequencies.
Step	1 LT: Shall send TST_BYTE_55 on IUT TXD. 2 UT: Shall measure the value of byte field sync preamble phase shifts at TP 1.
Iteration	Not applicable
Expected response	Step 2 The value of byte field sync preamble phase shifts measured at TP 1 complies with Table 4 = OK.
Remark	—

B.3.7 Length of byte field data bit phases

[Table B.11](#) defines the test structure of CTC_TX_CHARC_6.

Table B.11 — Test structure of CTC_TX_CHARC_6

Item	Content
CTC # – Title	1.CTC_TX_CHARC_6 – Length of byte field data bit phases
Purpose	This CTC verifies the correct length of byte field data bit phases.
Reference	Table 5

Table B.11 (continued)

Item	Content
CTC # - Title	1.CTC_TX_CHARC_6 – Length of byte field data bit phases
Prerequisite	1 LT with the capability to stimulate, reset and configure the IUT. 2 The UT is able to provide time measurement capabilities synchronised with the test step events (see Figure B.2).
Set-up	1 LT: Shall configure the IUT with default bit rate (see B.2.4.5.5). 2 LT: Shall configure the IUT with default working carrier frequencies (see Table B.4). 3 LT: Shall configure the IUT with default transmit signal number of carrier frequencies.
Step	1 LT: Shall send TST_BYTE_55 on IUT TXD. 2 UT: Shall measure the length of byte field data bit phases at TP 1.
Iteration	Not applicable
Expected response	Step 2 The length of byte field data bit phases measured at TP 1 complies with Table 5 = OK.
Remark	—

B.3.8 Value of byte field data bit phase shifts

[Table B.12](#) defines the test structure of CTC_TX_CHARC_7.

Table B.12 — Test structure of CTC_TX_CHARC_7

Item	Content
CTC # - Title	1.CTC_TX_CHARC_7 – Value of byte field data bit phase shifts
Purpose	This CTC verifies the correct value of byte field data bit phase shifts.
Reference	Table 5
Prerequisite	1 LT with the capability to stimulate, reset and configure the IUT. 2 The UT is able to provide time measurement capabilities synchronised with the test step events (see Figure B.2).
Set-up	1 LT: Shall configure the IUT with default bit rate (see B.2.4.5.5). 2 LT: Shall configure the IUT with default working carrier frequencies (see Table B.4). 3 LT: Shall configure the IUT with default transmit signal number of carrier frequencies.
Step	1 LT: Shall send TST_BYTE_55 on IUT TXD. 2 UT: Shall measure the value of byte field data bit phase shifts at TP 1. 3 LT: Shall send TST_BYTE_7A on IUT TXD. 4 UT: Shall measure the value of byte field data bit phase shifts at TP 1. 5 LT: Shall send TST_BYTE_65 on IUT TXD. 6 UT: Shall measure the value of byte field data bit phase shifts at TP 1. 7 LT: Shall send TST_BYTE_5F on IUT TXD. 8 UT: Shall measure the value of byte field data bit phase shifts at TP 1.
Iteration	Not applicable
Expected response	Step 2 The value of byte field data bit phase shifts measured at TP 1 complies with Table 5 = OK. Step 4 The value of byte field data bit phase shifts measured at TP 1 complies with Table 5 = OK. Step 6 The value of byte field data bit phase shifts measured at TP 1 complies with Table 5 = OK. Step 8 The value of byte field data bit phase shifts measured at TP 1 complies with Table 5 = OK.

Table B.12 (continued)

Item	Content
CTC # – Title	1.CTC_TX_CHARC_7 – Value of byte field data bit phase shifts
Remark	—

B.3.9 Length of break field phases

Table B.13 defines the test structure of CTC_TX_CHARC_8.

Table B.13 — Test structure of CTC_TX_CHARC_8

Item	Content
CTC # – Title	1.CTC_TX_CHARC_8 – Length of break field phases
Purpose	This CTC verifies the correct length of break field phases.
Reference	4.2.3
Prerequisite	1 LT with the capability to stimulate, reset and configure the IUT. 2 The UT is able to provide time measurement capabilities synchronised with the test step events (see Figure B.2).
Set-up	1 LT: Shall configure the IUT as master node. 2 LT: Shall configure the IUT with default bit rate (see B.2.4.5.5). 3 LT: Shall configure the IUT with default working carrier frequencies (see Table B.4). 4 LT: Shall configure the IUT with default transmit signal number of carrier frequencies.
Step	1 LT: Shall send TST_BREAK_FIELD on IUT TXD. 2 UT: Shall measure the length of break field phases at TP 1.
Iteration	Not applicable
Expected response	Step 2 The length of break field phases measured at TP 1 complies with as specified in 4.2.3 = OK.
Remark	—

B.3.10 Value of break field phase shifts

Table B.14 defines the test structure of CTC_TX_CHARC_9.

Table B.14 — Test structure of CTC_TX_CHARC_9

Item	Content
CTC # – Title	1.CTC_TX_CHARC_9 – Value of break field phase shifts.
Purpose	This CTC verifies the correct value of break field phase shifts.
Reference	4.2.3
Prerequisite	1 LT with the capability to stimulate, reset and configure the IUT. 2 The UT is able to provide time measurement capabilities synchronised with the test step events (see Figure B.2).
Set-up	1 LT: Shall configure the IUT as master node. 2 LT: Shall configure the IUT with default bit rate (see B.2.4.5.5). 3 LT: Shall configure the IUT with default working carrier frequencies (see Table B.4). 4 LT: Shall configure the IUT with default transmit signal number of carrier frequencies.
Step	1 LT: Shall send TST_BREAK_FIELD on IUT TXD. 2 UT: Shall measure the value of break field phase shifts at TP 1.
Iteration	Not applicable

Table B.14 (continued)

Item	Content
CTC # - Title	1.CTC_TX_CHARC_9 - Value of break field phase shifts.
Expected response	Step 2 The value of break field phase shifts measured at TP 1 complies with as specified in 4.2.3 = OK.
Remark	—

B.3.11 Length of wake-up signal phases

[Table B.15](#) defines the test structure of CTC_TX_CHARC_10.

Table B.15 — Test structure of CTC_TX_CHARC_10

Item	Content
CTC # - Title	1.CTC_TX_CHARC_10 - Length of wake-up signal phases
Purpose	This CTC verifies the correct length of wake-up signal phases.
Reference	4.2.4
Prerequisite	1 LT with the capability to stimulate, reset and configure the IUT. 2 The UT is able to provide time measurement capabilities synchronised with the test step events (see Figure B.2).
Set-up	1 LT: Shall configure the IUT with default bit rate (see B.2.4.5.5). 2 LT: Shall configure the IUT with default working carrier frequencies (see Table B.4). 3 LT: Shall configure the IUT with default transmit signal number of carrier frequencies.
Step	1 LT: Shall send TST_WAKEUP_SIGNAL_SHORT on IUT TXD. 2 UT: Shall measure the length of wake-up signal phases at TP 1. 3 LT: Shall send TST_WAKEUP_SIGNAL_LONG on IUT TXD. 4 UT: Shall measure the length of wake-up signal phases at TP 1. 5 LT: Shall send TST_BREAK_FIELD on IUT TXD. 6 UT: Shall measure the length of wake-up signal phases at TP 1.
Iteration	Not applicable
Expected response	Step 2 The length of wake-up signal phases measured at TP 1 complies with as specified in 4.2.4 = OK. Step 4 The length of wake-up signal phases measured at TP 1 complies with as specified in 4.2.4 = OK. Step 6 The length of wake-up signal phases measured at TP 1 complies with as specified in 4.2.4 = OK.
Remark	—

B.3.12 Value of wake-up signal phase shifts

[Table B.16](#) defines the test structure of CTC_TX_CHARC_11.

Table B.16 — Test structure of CTC_TX_CHARC_11

Item	Content
CTC # - Title	1.CTC_TX_CHARC_11 - Value of wake-up signal phase shifts
Purpose	This CTC verifies the correct value of wake-up signal phase shifts.
Reference	4.2.4
Prerequisite	1 LT with the capability to stimulate, reset and configure the IUT. 2 The UT is able to provide time measurement capabilities synchronised with the test step events (see Figure B.2).

Table B.16 (continued)

Item	Content
CTC # – Title	1.CTC_TX_CHARC_11 – Value of wake-up signal phase shifts
Set-up	1 LT: Shall configure the IUT with default bit rate (see B.2.4.5.5). 2 LT: Shall configure the IUT with default working carrier frequencies (see Table B.4). 3 LT: Shall configure the IUT with default transmit signal number of carrier frequencies.
Step	1 LT: Shall send TST_WAKEUP_SIGNAL_SHORT on IUT TXD. 2 UT: Shall measure the value of wake-up signal phase shifts at TP 1. 3 LT: Shall send TST_WAKEUP_SIGNAL_LONG on IUT TXD. 4 UT: Shall measure the value of wake-up signal phase shifts at TP 1. 5 LT: Shall send TST_BREAK_FIELD on IUT TXD. 6 UT: Shall measure the value of wake-up signal phase shifts at TP 1.
Iteration	Not applicable
Expected response	Step 2 The value of wake-up signal phase shifts measured at TP 1 complies with as specified in 4.2.4 = OK. Step 4 The value of wake-up signal phase shifts measured at TP 1 complies with as specified in 4.2.4 = OK. Step 6 The value of wake-up signal phase shifts measured at TP 1 complies with as specified in 4.2.4 = OK.
Remark	—

B.3.13 Transmitter bit rate deviation

[Table B.17](#) defines the test structure of CTC_TX_CHARC_12.

Table B.17 — Test structure of CTC_TX_CHARC_12

Item	Content
CTC # – Title	1.CTC_TX_CHARC_12 – Transmitter bit rate deviation
Purpose	This CTC verifies the correct bit rate deviation of a DC-LIN EPL transmitter.
Reference	Table 6
Prerequisite	1 LT with the capability to stimulate, reset and configure the IUT. 2 The UT is able to provide time measurement capabilities synchronised with the test step events (see Figure B.2).
Set-up	1 LT: Shall configure the IUT with default bit rate (see B.2.4.5.5). 2 LT: Shall configure the IUT with default working carrier frequencies (see Table B.4). 3 LT: Shall configure the IUT with default transmit signal number of carrier frequencies.
Step	1 LT: Shall send TST_BYTE_55 on IUT TXD. 2 UT: Shall measure the overall transmission time measured at TP 1. 3 UT: Shall measure the t_{BIT} on RXD at TP 3. 4 LT: Shall send TST_BYTE_7A on IUT TXD. 5 UT: Shall measure the overall transmission time measured at TP 1. 6 UT: Shall measure the t_{BIT} on RXD at TP 3. 7 LT: Shall send TST_BYTE_65 on IUT TXD.
	8 UT: Shall measure the overall transmission time measured at TP 1. 9 UT: Shall measure the t_{BIT} on RXD at TP 3. 10 LT: Shall send TST_BYTE_5F on IUT TXD. 11 UT: Shall measure the overall transmission time measured at TP 1. 12 UT: Shall measure the t_{BIT} on RXD at TP 3.

Table B.17 (continued)

Item	Content
CTC # - Title	1.CTC_TX_CHARC_12 - Transmitter bit rate deviation
Iteration	13 LT: Shall configure the IUT bit rate to BR_9_6K. REPEAT step 1 to step 12; 1 time REPEAT END 14 LT: Shall configure the IUT bit rate to BR_10K. REPEAT step 1 to step 12; 1 time REPEAT END
Expected response	Step 2 The overall transmission time measured at TP 1 should be less than $\pm 0,5$ % from the selected bit rate, as specified in Table 6 = OK. Step 3 The t_{BIT} measured on RXD at TP 3 should be less than $\pm 0,5$ % from the selected bit rate, as specified in Table 6 = OK. Step 5 The overall transmission time measured at TP 1 should be less than $\pm 0,5$ % from the selected bit rate, as specified in Table 6 = OK. Step 6 The t_{BIT} measured on RXD at TP 3 should be less than $\pm 0,5$ % from the selected bit rate, as specified in Table 6 = OK. Step 8 The overall transmission time measured at TP 1 should be less than $\pm 0,5$ % from the selected bit rate, as specified in Table 6 = OK. Step 9 The t_{BIT} measured on RXD at TP 3 should be less than $\pm 0,5$ % from the selected bit rate, as specified in Table 6 = OK. Step 11 The overall transmission time measured at TP 1 should be less than $\pm 0,5$ % from the selected bit rate, as specified in Table 6 = OK. Step 12 The t_{BIT} measured on RXD at TP 3 should be less than $\pm 0,5$ % from the selected bit rate, as specified in Table 6 = OK.
Remark	—

B.3.14 Line-out monitoring

[Table B.18](#) defines the test structure of CTC_TX_CHARC_13.

Table B.18 — Test structure of CTC_TX_CHARC_13

Item	Content
CTC # - Title	1.CTC_TX_CHARC_13 - Line-out monitoring
Purpose	This CTC verifies the line-out monitoring of a DC-LIN EPL.
Reference	Table 10 and Table 12
Prerequisite	1 LT with the capability to stimulate, reset and configure the IUT. 2 The UT is able to provide time measurement capabilities synchronised with the test step events (see Figure B.2). 3 The UT is able to provide attenuation capabilities of line-out level, fulfilling the following conditions (see Table 10): — $L_{out_lev} \geq L_{out_thr_lev}$ — $L_{out_lev} < L_{out_thr_lev}$
Set-up	1 LT: Shall configure the IUT with default bit rate (see B.2.4.5.5). 2 LT: Shall configure the IUT with default working carrier frequencies (see Table B.4). 3 LT: Shall configure the IUT with CF_{max} transmit signal number of carrier frequencies (i.e. 1, 2, 3, or 4).
Step	1 LT: Shall send TST_BYTE_55 on IUT TXD.

Table B.18 (continued)

Item	Content
CTC # – Title	1.CTC_TX_CHARC_13 – Line-out monitoring
	2 UT: Shall keep line-out level complies with $L_{out_lev} \geq L_{out_thr_lev}$ for the entire duration of TST_BYTE_55 transmission. 3 UT: Shall observe RXD at TP 3 and TXD at TP 4 for the entire duration of TST_BYTE_55 transmission. 4 LT: Shall send TST_BYTE_55 on IUT TXD. 5 UT: Shall keep line-out level complies with $L_{out_lev} \geq L_{out_thr_lev}$ until TST_BYTE_55 data bit $n = 0$ transmission is completed. Then, UT shall attenuate the line-out level to comply with $L_{out_lev} < L_{out_thr_lev}$ for the remaining duration of TST_BYTE_55 transmission. 6 UT: Shall observe RXD at TP 3 and TXD at TP 4 for the entire duration of TST_BYTE_55 transmission. 7 UT: Shall measure line-out phase shifts at TP 1 for the entire duration of TST_BYTE_55 transmission.
Iteration	REPEAT step 4; $k = 1$ to $k = 7$ times 9 UT: Shall keep line-out level complies with $L_{out_lev} \geq L_{out_thr_lev}$ until TST_BYTE_55 data bit $n = k$ transmission is completed. Then, UT shall attenuate the line-out level to comply with $L_{out_lev} < L_{out_thr_lev}$ for the remaining duration of TST_BYTE_55 transmission. 10 UT: Shall observe RXD at TP 3 and TXD at TP 4 for the entire duration of TST_BYTE_55 transmission. 11 UT: Shall measure line-out phase shifts at TP 1 for the entire duration of TST_BYTE_55 transmission. REPEAT END
Expected response	Step 3 The RXD observed at TP 3 should be equal to TXD observed at TP 4 for the entire duration of TST_BYTE_55 transmission, as specified in Table 12 = OK. Step 6 The RXD observed at TP 3 should be equal to TXD observed at TP 4 until $L_{out_err_cond}$ is triggered active = OK. Then, from $t_{l_out_mon_samp}$ of data bit $n = 1$ to $n = 8$, RXD should be equal to TXD for the remaining duration of the TST_BYTE_55 transmission (including stop bit frame error) = OK. Step 7 The IUT line-out should be measured with constant phase transmission (i.e. no phase shift) from data bit $n = 1$ to $n = 8$, as specified in Table 12, and shown in Figure 18 = OK. Step 10 The RXD observed at TP 3 should be equal to TXD observed at TP 4 until $L_{out_err_cond}$ is triggered active = OK. Then, from $t_{l_out_mon_samp}$ of data bit $n = k + 1$ to $n = 8$, RXD should be equal to TXD for the remaining duration of the TST_BYTE_55 transmission (including stop bit frame error) = OK. Step 11 The IUT line-out should be measured with constant phase transmission (i.e. no phase shift) from data bit $n = k + 1$ to $n = 8$, as specified in Table 12, and shown in Figure 18 = OK.
Remark	—

B.3.15 Line-out output voltage

[Table B.19](#) defines the test structure of CTC_TX_CHARC_14.

Table B.19 — Test structure of CTC_TX_CHARC_14

Item	Content
CTC # – Title	1.CTC_TX_CHARC_14 – Line-out output voltage
Purpose	This CTC verifies the value of a DC-LIN EPL line-out output voltage.
Reference	Table 9

Table B.19 (continued)

Item	Content
CTC # - Title	1.CTC_TX_CHARC_14 – Line-out output voltage
Prerequisite	1 LT with the capability to stimulate, reset and configure the IUT. 2 The UT is able to provide time measurement capabilities synchronised with the test step events (see Figure B.2). 3 The LT is able to activate the IUT low-power mode.
Set-up	1 LT: Shall configure the IUT with default bit rate (see B.2.4.5.5). 2 LT: Shall configure the IUT with default working carrier frequencies (see Table B.4). 3 LT: Shall configure the IUT with default transmit signal number of carrier frequencies.
Step	1 LT: Shall send TST_BYTE_55 on IUT TXD (i.e. T_{tx_mode} is active). 2 UT: Shall measure the IUT line-out voltage level measured at TP 1. 3 LT: Shall set IUT TXD to '1' (i.e. IUT T_{tx_mode} is inactive). 4 UT: Shall measure the IUT line-out voltage level measured at TP 1. 5 LT: Shall activate IUT low-power mode. 6 UT: Shall measure the IUT line-out voltage level measured at TP 1.
Iteration	Not applicable
Expected response	Step 2 The IUT line-out voltage level measured at TP 1, should be in the range of 0,96 Vpp and 1,44 Vpp, as specified in Table 9 = OK. Step 4 The IUT line-out voltage level measured at TP 1, should be in the range of 0 Vpp and 0,1 Vpp, as specified in Table 9 = OK. Step 6 The IUT line-out voltage level measured at TP 1, should be in the range of 0 Vpp and 0,1 Vpp, as specified in Table 9 = OK.
Remark	—

B.3.16 Line-out input impedance

[Table B.20](#) defines the test structure of CTC_TX_CHARC_15.

Table B.20 — Test structure of CTC_TX_CHARC_15

Item	Content
CTC # - Title	1.CTC_TX_CHARC_15 – Line-out input impedance
Purpose	This CTC verifies the minimum line-out input impedance of a DC-LIN EPL.
Reference	Table 9
Prerequisite	1 LT with the capability to stimulate, reset and configure the IUT. 2 The UT is able to provide time measurement capabilities synchronised with the test step events (see Figure B.2). 3 The LT is able to activate the IUT low-power mode.
Set-up	—
Step	1 LT: Shall set IUT TXD to '1' (i.e. IUT T_{tx_mode} is inactive). 2 UT: Shall measure the IUT line-out input impedance. 3 LT: Shall activate IUT low-power mode. 4 UT: Shall measure the IUT line-out input impedance.
Iteration	Not applicable
Expected response	Step 2 The measured IUT line-out input impedance should be ≥ 100 kΩ, as specified in Table 9 = OK. Step 4 The measured IUT line-out input impedance should be ≥ 100 kΩ, as specified in Table 9 = OK.

Table B.20 (continued)

Item	Content
CTC # - Title	1.CTC_TX_CHARC_15 - Line-out input impedance
Remark	—

B.4 Receiver characteristics timing

B.4.1 General

This subclause verifies if the DC-LIN EPL receiver characteristics timing of the IUT as slave node or master node conform to the receiver characteristics timing parameters that are specified in this document. The IUT line-in is stimulated by a test pattern transmit signal injected onto the powerline (lower tester, see Table B.2, and Figure B.3). The resulting behaviour is observed and compared with the expected behaviour (upper tester).

Figure B.3 depicts the test configuration of the test system 'Receiver characteristics timing'.

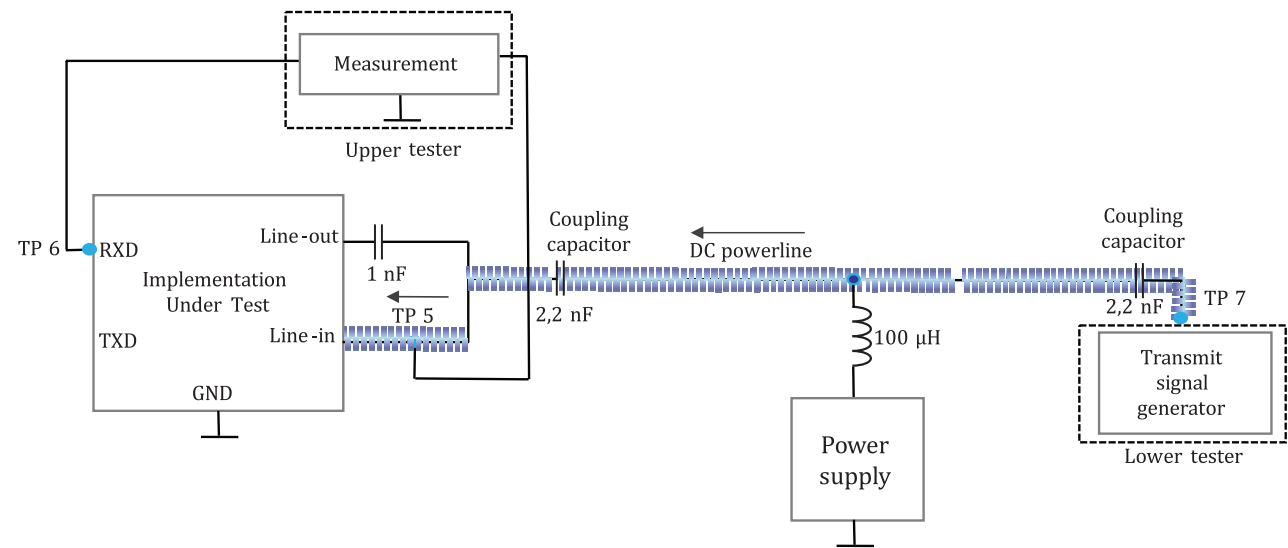


Figure B.3 — Test system: Receiver characteristics timing

B.4.2 Line-in input impedance

Table B.21 defines the test structure of CTC_RX_CHARC_1.

Table B.21 — Test structure of CTC_RX_CHARC_1

Item	Content
CTC # - Title	1.CTC_RX_CHARC_1 - Line-in input impedance
Purpose	This CTC verifies the minimum line-in input impedance of a DC-LIN EPL.
Reference	Table 9
Prerequisite	1 LT with the capability to stimulate, reset and configure the IUT. 2 The UT is able to provide time measurement capabilities synchronised with the test step events (see Figure B.3). 3 The LT is able to activate the IUT low-power mode.

Table B.21 (continued)

Item	Content
CTC # - Title	1.CTC_RX_CHARC_1 - Line-in input impedance
Set-up	1 LT: Shall configure the IUT with default bit rate (see B.2.4.5.5). 2 LT: Shall configure the IUT with default working carrier frequencies (see Table B.4). 3 LT: Shall configure the IUT with default transmit signal number of carrier frequencies.
Step	1 LT: Shall send TST_BYTE_55 on IUT TXD (i.e. T_{tx_mode} is active). 2 UT: Shall measure the IUT line-in input impedance at TP 5. 3 LT: Shall set IUT TXD to '1' (i.e. IUT T_{tx_mode} is inactive). 4 UT: Shall measure the IUT line-in input impedance at TP 5. 5 LT: Shall activate IUT low-power mode. 6 UT: Shall measure the IUT line-in input impedance at TP 5.
Iteration	Not applicable
Expected response	Step 2 The measured IUT line-in input impedance at TP 5, should be $\geq 5 \text{ k}\Omega$, as specified in Table 9 = OK. Step 4 The measured IUT line-in input impedance at TP 5, should be $\geq 5 \text{ k}\Omega$, as specified in Table 9 = OK. Step 6 The measured IUT line-in input impedance at TP 5, should be $\geq 5 \text{ k}\Omega$, as specified in Table 9 = OK.
Remark	—

B.4.3 Line-in minimum input voltage

[Table B.22](#) defines the test structure of CTC_R $\times$ _CHARC_2.

Table B.22 — Test structure of CTC_RX_CHARC_2

Item	Content
CTC # - Title	1.CTC_RX_CHARC_2 - Line-in minimum input voltage
Purpose	This CTC verifies a DC-LIN EPL proper reception of a byte field when minimum line-in input voltage is applied.
Reference	Table 8 , and Table 9
Prerequisite	1 LT with the capability to stimulate, reset and configure the IUT. 2 The UT is able to provide time measurement capabilities synchronised with the test step events (see Figure B.3). 3 The UT is able to provide attenuation capabilities of line-in level to 0,005 Vpp.
Set-up	1 LT: Shall configure the IUT with default bit rate (see B.2.4.5.5). 2 LT: Shall configure the IUT with default working carrier frequencies (see Table B.4). 3 LT: Shall configure the IUT with CF_{max} transmit signal number of carrier frequencies (i.e. 1, 2, 3, or 4).
Step	1 LT: Shall generate TXS_S_BYTE_55 on TP 7 and attenuate the IUT line-in voltage level to 0,005 Vpp. 2 UT: Shall observe IUT RXD at TP 6.
Iteration	Not applicable
Expected response	Step 2 The IUT RXD observed at TP 6 should show TST_BYTE_55, as specified in Table 8 = OK.
Remark	—

B.4.4 Byte field/s reception

[Table B.23](#) defines the test structure of CTC_RX_CHARC_3.

Table B.23 — Test structure of CTC_RX_CHARC_3

Item	Content
CTC # – Title	1.CTC_RX_CHARC_3 – Byte field/s reception
Purpose	This CTC verifies a DC-LIN EPL proper reception of a byte field/s that complies with t_{RX_delay} .
Reference	4.3.6
Prerequisite	1 LT with the capability to stimulate, reset and configure the IUT. 2 The UT is able to provide time measurement capabilities synchronised with the test step events (see Figure B.3).
Set-up	1 LT: Shall configure the IUT with default bit rate (see B.2.4.5.5). 2 LT: Shall configure the IUT with default working carrier frequencies (see Table B.4). 3 LT: Shall configure the IUT with CF_{max} transmit signal number of carrier frequencies (i.e. 1, 2, 3, or 4).
Step	1 LT: Shall generate TXS_S_BYTE_55 on TP 7. 2 UT: Shall observe IUT RXD at TP 6. 3 LT: Shall generate TXS_BYTE_7A_CONSEC_55 on TP 7. 4 UT: Shall observe IUT RXD at TP 6. 5 LT: Shall generate TXS_BYTE_55_CONSEC_55_CONSEC_55 on TP 7. 6 UT: Shall observe IUT RXD at TP 6.
Iteration	Not applicable
Expected response	Step 2 The IUT RXD observed at TP 6 should show a byte field start bit falling edge within t_{RX_delay} with accordance to the start time of the generated test pattern (t_{SB_TX}) (see Figure 13) = OK. Step 4 The IUT RXD observed at TP 6 should show a byte field start bit falling edge within t_{RX_delay} with accordance to the start time of the generated test pattern (t_{SB_TX}) (see Figure 13) = OK. Step 6 The IUT RXD observed at TP 6 should show a byte field start bit falling edge within t_{RX_delay} with accordance to the start time of the generated test pattern (t_{SB_TX}) (see Figure 13) = OK.
Remark	—

B.4.5 Break field reception

[Table B.24](#) defines the test structure of CTC_RX_CHARC_4.

Table B.24 — Test structure of CTC_RX_CHARC_4

Item	Content
CTC # – Title	1.CTC_RX_CHARC_4 – Break field reception
Purpose	This CTC verifies a DC-LIN EPL proper break field reception.
Reference	Table 8
Prerequisite	1 LT with the capability to stimulate, reset and configure the IUT. 2 The UT is able to provide time measurement capabilities synchronised with the test step events (see Figure B.3).
Set-up	1 LT: Shall configure the IUT as slave node. 2 LT: Shall configure the IUT with default bit rate (see B.2.4.5.5). 3 LT: Shall configure the IUT with default working carrier frequencies (see Table B.4). 4 LT: Shall configure the IUT with CF_{max} transmit signal number of carrier frequencies (i.e. 1, 2, 3, or 4).
Step	1 LT: Shall generate TXS_BREAK_FIELD on TP 7. 2 UT: Shall observe IUT RXD at TP 6.

Table B.24 (continued)

Item	Content
CTC # - Title	1.CTC_RX_CHARC_4 - Break field reception
Iteration	Not applicable
Expected response	Step 2 The IUT RXD observed at TP 6 should show TST_BREAK_FIELD, as specified in Table 8 = OK.
Remark	—

B.4.6 Wake-up signal reception

[Table B.25](#) defines the test structure of CTC_RX_CHARC_5.

Table B.25 — Test structure of CTC_RX_CHARC_5

Item	Content
CTC # - Title	1.CTC_RX_CHARC_5 - Wake-up signal reception
Purpose	This CTC verifies a DC-LIN EPL proper wake-up signal reception.
Reference	Table 8
Prerequisite	1 LT with the capability to stimulate, reset and configure the IUT. 2 The UT is able to provide time measurement capabilities synchronised with the test step events (see Figure B.3).
Set-up	1 LT: Shall configure the IUT with default bit rate (see B.2.4.5.5). 2 LT: Shall configure the IUT with default working carrier frequencies (see Table B.4). 3 LT: Shall configure the IUT with $CF_{\max}$ transmit signal number of carrier frequencies (i.e. 1, 2, 3, or 4).
Step	1 LT: Shall generate TXS_WAKEUP_SIGNAL_SHORT on TP 7. 2 UT: Shall observe IUT RXD at TP 6. 3 LT: Shall generate TXS_WAKEUP_SIGNAL_LONG on TP 7. 4 UT: Shall observe IUT RXD at TP 6. 5 LT: Shall generate TXS_BREAK_FIELD on TP 7. 6 UT: Shall observe IUT RXD at TP 6.
Iteration	Not applicable
Expected response	Step 2 The IUT RXD observed at TP 6 should show TST_WAKEUP_SIGNAL_SHORT, as specified in Table 8 = OK. Step 4 The IUT RXD observed at TP 6 should show TST_WAKEUP_SIGNAL_LONG, as specified in Table 8 = OK. Step 6 The IUT RXD observed at TP 6 should show TST_BREAK_FIELD, as specified in Table 8 = OK.
Remark	—

B.4.7 Reception error condition

[Table B.26](#) defines the test structure of CTC_RX_CHARC_6.

Table B.26 — Test structure of CTC_RX_CHARC_6

Item	Content
CTC # - Title	1.CTC_RX_CHARC_6 - Reception error condition
Purpose	This CTC verifies a proper detection of DC-LIN EPL active reception error condition (RX _{err_cond}).
Reference	Table 8

Table B.26 (continued)

Item	Content
CTC # – Title	1.CTC_RX_CHARC_6 – Reception error condition
Prerequisite	1 LT with the capability to stimulate, reset and configure the IUT. 2 The UT is able to provide time measurement capabilities synchronised with the test step events (see Figure B.3).
Set-up	1 LT: Shall configure the IUT with default bit rate (see B.2.4.5.5). 2 LT: Shall configure the IUT with default working carrier frequencies (see Table B.4). 3 LT: Shall configure the IUT with default transmit signal number of carrier frequencies.
Step	1 LT: Shall generate TXS_S_BYTE_55_ERR_COND_N2 on TP 7. 2 UT: Shall observe IUT RXD at TP 6. 3 LT: Shall generate TXS_S_BYTE_55_ERR_COND_N4 on TP 7. 4 UT: Shall observe IUT RXD at TP 6. 5 LT: Shall generate TXS_S_BYTE_55_ERR_COND_N6 on TP 7. 6 UT: Shall observe IUT RXD at TP 6.
Iteration	Not applicable
Expected response	Step 2 The IUT RXD observed at TP 6 should indicate a stop bit frame error, as specified in Table 8 , and shown in Figure 18 = OK. Step 4 The IUT RXD observed at TP 6 should indicate a stop bit frame error, as specified in Table 8 , and shown in Figure 18 = OK. Step 6 The IUT RXD observed at TP 6 should indicate a stop bit frame error, as specified in Table 8 , and shown in Figure 18 = OK.
Remark	—

B.4.8 Receiver bit rate deviation

[Table B.27](#) defines the test structure of CTC_RX_CHARC_7.

Table B.27 — Test structure of CTC_RX_CHARC_7

Item	Content
CTC # – Title	1.CTC_RX_CHARC_7– Receiver bit rate deviation
Purpose	This CTC verifies the correct bit rate deviation of a DC-LIN EPL receiver.
Reference	Table 6
Prerequisite	1 LT with the capability to stimulate, reset and configure the IUT. 2 The UT is able to provide time measurement capabilities synchronised with the test step events (see Figure B.3).
Set-up	1 LT: Shall configure the IUT with default bit rate (see B.2.4.5.5). 2 LT: Shall configure the IUT with default working carrier frequencies (see Table B.4). 3 LT: Shall configure the IUT with default transmit signal number of carrier frequencies.
Step	1 LT: Shall generate TXS_S_BYTE_55 on TP 7. 2 UT: Shall measure the t_{BIT} on RXD at TP 6. 3 LT: Shall generate TXS_S_BYTE_7A on TP 7. 4 UT: Shall measure the t_{BIT} on RXD at TP 6. 5 LT: Shall generate TXS_S_BYTE_65 on TP 7. 6 UT: Shall measure the t_{BIT} on RXD at TP 6. 7 LT: Shall generate TXS_S_BYTE_5F on TP 7. 8 UT: Shall measure the t_{BIT} on RXD at TP 6.

Table B.27 (continued)

Item	Content
CTC # – Title	1.CTC_RX_CHARC_7– Receiver bit rate deviation
Iteration	9 UT: Shall configure the IUT bit rate to BR_9_6K. REPEAT step 1 to step 8; 1 time REPEAT END 10 UT: Shall configure the IUT bit rate to BR_10K. REPEAT step 1 to step 8; 1 time REPEAT END
Expected response	Step 2 The t_{BIT} measured on RXD at TP 6 should be less than $\pm 0,5$ % from the selected bit rate, as specified in Table 6 = OK. Step 4 The t_{BIT} measured on RXD at TP 6 should be less than $\pm 0,5$ % from the selected bit rate, as specified in Table 6 = OK. Step 6 The t_{BIT} measured on RXD at TP 6 should be less than $\pm 0,5$ % from the selected bit rate, as specified in Table 6 = OK. Step 8 The t_{BIT} measured on RXD at TP 6 should be less than $\pm 0,5$ % from the selected bit rate, as specified in Table 6 = OK.
Remark	—

B.5 DC-LIN EPL compliance to ISO 17987-6

Table B.28 specifies ISO 17987-6 test cases, which are either not applicable as they are not compliant with this document or require modification. The remaining of test cases specified in ISO 17987-6 shall apply.

Table B.28 — DC-LIN EPL non-compliant test cases with ISO 17987-6

ISO 17987-6 Test case	Short description	Modification/change	Reason for modification/change
[PT-CT 23]	Sample point test, IUT as slave node	Exclude	DC-LIN EPL doesn't support 1/16 bit time.
[PT-CT 15]	Bit rate tolerance, IUT as master node	Exclude for class C specified in ISO 17987-6	Test is not applicable for DC-LIN EPL interface itself; part of TXD/RXD interface
[PT-CT 16]	Bit rate tolerance, IUT as slave node without making use of synchronization	Exclude for class C specified in ISO 17987-6	Test is not applicable for DC-LIN EPL interface itself; part of TXD/RXD interface
[PT-CT 17]	Bit rate tolerance, IUT as slave node with making use of synchronization	Exclude for class C specified in ISO 17987-6	Test is not applicable for DC-LIN EPL interface itself; part of TXD/RXD interface
[PT-CT 5]	Variation of length of break field low phase, IUT as slave node	a) Change t_{BIT} to 1 to 9 b) Change t_{BIT} to 11 to 26	DC-LIN EPL quantization of 1 t_{BIT}
[PT-CT 7] .1	VarVariation of length of break delimiter, IUT as slave node	Change class C 0,75 t_{BIT} to 1 t_{BIT}	DC-LIN EPL quantization of 1 t_{BIT}
[PT-CT 7] .2	VarVariation of length of break delimiter, IUT as slave node	Change class C 14,6 t_{BIT} to 14 t_{BIT}	DC-LIN EPL quantization of 1 t_{BIT}

Bibliography

- [1] ISO 7498-1, *Information processing systems — Open systems interconnection — Basic reference model*
- [2] ISO/IEC 10731, *Information technology — Open Systems Interconnection — Basic Reference Model — Conventions for the definition of OSI services*
- [3] ISO 17987-2:2016, *Road vehicles — Local Interconnect Network (LIN) — Part 2: Transport protocol and network layer services*
- [4] ISO 17987-3, *Road vehicles — Local Interconnect Network (LIN) — Part 3: Protocol specification*
- [5] ISO/TR 17987-5, *Road vehicles — Local Interconnect Network (LIN) — Part 5: Application programmers interface (API)*
- [6] ISO/IEC 10731, *Information technology — Open Systems Interconnection — Basic Reference Model — Conventions for the definition of OSI services*

.....